

CHAPTER 5

CHIMNEYS AND VENTS

SECTION FGC 501 GENERAL

501.1 Scope. This chapter shall govern the installation, alteration, maintenance, design, minimum safety requirements, repair and approval of factory-built chimneys, chimney liners, vents and connectors, field-built chimneys and connectors and the utilization of masonry chimneys serving gas-fired appliances. The requirements for the installation, maintenance, repair and approval of factory-built chimneys, chimney liners, vents and connectors serving appliances burning fuels other than fuel gas shall be regulated by the *New York City Mechanical Code*. The construction, repair, maintenance and approval of masonry chimneys shall be regulated by the *New York City Building Code*. Venting systems shall be designed in accordance with this chapter and comply with the requirements of the *New York City Air Pollution Control Code*.

501.1.1 Existing chimneys and vents. Existing chimneys and vents shall comply with the requirements of Section 28-104.13 of the *Administrative Code* and Sections 501.1.1.1 through 501.1.1.8 of this code.

501.1.1.1 Responsibility of owner of taller building. Whenever a building is erected, enlarged, or increased in height so that any portion of such building, except chimneys or vents, extends higher than the top of any existing chimneys or vents within 100 feet (30 480 mm), the owner of such new or altered building shall have the responsibility of altering such chimneys or vents to make them conform with the requirements of this chapter. A chimney or vent that is no longer connected with a fireplace or combustion or other equipment for which a chimney or vent was required shall be exempt from this requirement. Such alterations shall be accomplished by one of the following means or a combination thereof:

1. Carry up the existing chimneys or vents to the height required in this chapter.
2. Offset such chimneys or vents to a distance beyond that required by this chapter from the new or altered building provided that the new location of the outlet of the offset chimney or vent shall otherwise comply with the requirements of this chapter.

Such requirements shall not dispense with or modify any additional requirements that may be applicable pursuant to rules of the *New York City Department of Environmental Protection*.

501.1.1.1.1 Chimney and vent plan. Applications for a new or altered building shall include a chimney and vent plan submitted pursuant to Section 107.18 of the *New York City Building Code*.

501.1.1.1.2 Protection of draft. After the alteration of a chimney or vent as required by this section, it shall be the responsibility of the owner of the new or altered building to provide any mechanical equipment or devices necessary to maintain the proper draft in the equipment.

501.1.1.1.3 Written notification, plans and required documents. The owner of the new or altered building shall notify the owner of any building that may require a chimney or vent to be altered. Notification, plans and required documents shall comply with the requirements of Sections 501.1.1.3.1 through 501.1.1.3.3.

501.1.1.3.1 First notice. Written notice in a form acceptable to the department shall be provided to the building owner not less than 60 days prior to a request for permit for construction on the new or altered building. Such notice shall include a request for access to determine the need to alter the existing chimney or vent and a description of such work. Notice shall be sent by regular mail and certified mail, return receipt requested. A copy of such return receipt shall be filed with the department.

501.1.1.3.2 Second notice. Written notice in a form acceptable to the department shall be provided to the building owner not more than 45 days following commencement of work after a permit has been issued for the new or altered building. Such notice shall include a request for access to determine the need to alter the existing chimney or vent and a description of such work. Notice shall be sent by regular mail and certified mail, return receipt requested. The second notice shall also be posted by a licensed process server at the public entrance of the building requiring a chimney or vent to be altered. A copy of such return receipt and proof of service by the licensed process server shall be filed with the department.

Exceptions:

1. A second notice shall not be required where an application to alter the affected chimney or vent has been filed with the department.
2. A second notice shall not be required where access is granted and conditions are observed that result in a determination that chimney or vent alteration is not required and a revised chimney and vent plan is submitted to the department.

501.1.1.3.3 Plans and required documentation for alteration work. Where access is granted and conditions are observed that result in a determination that chimney or vent alteration is required, plans for such alteration work shall be provided to the owner of the existing building and a request for written consent to submit construction documents and perform such work shall be made.

501.1.1.4 Approval. The construction documents for the proposed chimney extension, alteration or relocation shall be submitted to the department pursuant to Section 28-104 of the *Administrative Code*. No certificate of occupancy shall be issued for the new building pursuant to Section 28-118.23 of the *Administrative Code* until the work associated with the construction documents for the proposed chimney extension, alteration or relocation has been signed-off by the department.

Exceptions:

1. A certificate of occupancy may be issued where access is granted and conditions are observed that result in a determination that chimney or vent alteration is not required and a revised chimney or vent plan is submitted pursuant to Section 107.18 of the *New York City Building Code* documenting such.
2. A certificate of occupancy may be issued in accordance with Section 28-118.23, Exception 2 of the *Administrative Code*.

501.1.1.5 Refusal of consent. If consent is not granted by the owner of the affected building to do the alteration work required by this section, such owner shall signify his or her refusal in writing to the owner of the new or altered building and to the commissioner; and the owner of the new or altered building having provided the notices required by Section 501.1.1.3 shall thereupon be released from any responsibility for the proper operation of the equipment due to loss of draft and for any health hazard or nuisance that may occur as a result of the new or altered building. Such responsibilities shall then be assumed by the owner of the previously constructed building. Similarly, should such owner fail to grant consent within 45 days from the date of the second notice or fail to signify his or her refusal, he or she shall then assume all responsibilities as prescribed above.

501.1.1.6 Procedure. It shall be the obligation of the owner of the new or altered building to:

1. Prepare and submit a chimney and vent plan to the department pursuant to Section 107.18 of the *New York City Building Code*.
2. Provide required notification pursuant to Section 501.1.1.3 of this code.
3. Provide plans pursuant to Section 501.1.1.3.3 of this code.
4. Prepare and submit construction documents to the department pursuant to Section 28-104 of the *Administrative Code* for the alteration of existing chimneys or vents which conform to the requirements of this chapter;
5. Obtain permit(s) for the proposed work in accordance with Section 28-105 of the *Administrative Code*;
6. Schedule this work so as to create a minimum of disturbance to the occupants of the affected building;
7. Provide such essential services as are normally supplied by the equipment while it is out of service;
8. Where necessary, support such extended chimneys, vents and equipment from this building or to carry up such chimneys or vents within his or her building;
9. Provide for the maintenance, repair, and/or replacement of such extensions and added equipment;
10. Make such alterations of the same material as the original chimney or vent so as to maintain the same quality and appearance, except where the owner of the chimney or vent shall give his or her consent to do otherwise. All work shall be done in such fashion as to maintain the architectural aesthetics of the existing building. Where there is practical difficulty in complying strictly with the provisions of this item, the commissioner may permit an equally safe alternative;
11. Comply with the tenant protection plan requirements of Section 28-120 of the *Administrative Code*; and
12. Comply with inspection and sign-off requirements of Section 28-116 of the *Administrative Code*.

501.1.1.7 Existing violations. Any existing violations on the previously constructed equipment shall be corrected by the owner of the equipment before any equipment is added or alterations made at the expense of the owner of the new or altered building.

501.1.1.8 Variance. The commissioner may grant a variance in accordance with the provisions of this code.

501.2 General. Every appliance shall discharge the products of combustion to the outdoors, except for appliances exempted by Section 501.8.

Exception: Commercial cooking appliances vented by a Type I hood installed in accordance with Section 507 of the *New York City Mechanical Code*.

501.2.1 Design. Chimneys and vents shall be designed and constructed so as to provide the necessary draft and capacity for each appliance connected to completely exhaust the products of combustion to the outside air. The temperature on adjacent combustible surfaces shall not be raised above 160°F (71.1°C). Chimneys and vents shall be designed to resist the effects of condensation that would cause deterioration of the chimney or vent.

501.2.2 Outlets. The outlet shall be arranged so that the flue gases are not directed so that they jeopardize people, overheat combustible structures, or enter building openings in the vicinity of the outlet. Gas-fired appliances shall be vented in accordance with this code and NFPA 54.

501.2.3 Support. Chimneys and vents shall not be supported by the equipment they serve unless such equipment has been specifically designed for such loads.

501.2.4 Changes in appliance fuels. Masonry chimneys shall be constructed in accordance with Section 507.

501.2.5 Exhaust gases from internal combustion engines and turbines. Exhaust pipes shall be constructed in accordance with Section 508.

501.3 Masonry chimneys. Masonry chimneys shall be constructed in accordance with Section 503.5.3 and the *New York City Building Code*.

501.4 Minimum size of chimney or vent. Chimneys and vents shall be sized in accordance with Sections 503 and 504.

501.5 Abandoned inlet openings. Abandoned inlet openings in chimneys and vents shall be closed by an approved method.

501.6 Positive pressure. Where an appliance equipped with a mechanical forced draft system creates a positive pressure in the venting system, the venting system shall be designed for positive pressure applications.

501.7 Connection to fireplace. Connection of appliances to chimney flues serving fireplaces shall be in accordance with Sections 501.7.1 through 501.7.3.

501.7.1 Closure and access. A noncombustible seal shall be provided below the point of connection to prevent entry of room air into the flue. Means shall be provided for access to the flue for inspection and cleaning.

501.7.2 Connection to factory-built fireplace flue. An appliance shall not be connected to a flue serving a factory-built fireplace unless the appliance is specifically listed for such installation. The connection shall be made in accordance with the appliance manufacturer's instructions.

501.7.3 Connection to masonry fireplace flue. A connector shall extend from the appliance to the flue serving a masonry fireplace such that the flue gases are exhausted directly into the flue. The connector shall be accessible or removable for inspection and cleaning of both the connector and the flue. Listed direct connection devices shall be installed in accordance with their listing.

501.8 Appliances not required to be vented. The following appliances shall not be required to be vented:

1. Ranges.
2. Built-in domestic cooking units listed and marked for optional venting.
3. Hot plates and laundry stoves.
4. Type 1 clothes dryers (Type 1 clothes dryers shall be exhausted in accordance with the requirements of Section 504 of the *New York City Mechanical Code*).
5. A single booster-type automatic instantaneous water heater, where designed and used solely for the sanitizing rinse requirements of a dishwashing machine, provided that the heater is installed in a commercial kitchen having a mechanical exhaust system. Where installed in this manner, the draft hood, if required, shall be in place and unaltered and the draft hood outlet shall be not less than 36 inches (914.4 mm) vertically and 6 inches (152.4 mm) horizontally from any surface other than the heater.
6. Refrigerators.
7. Counter appliances.
8. Specialized appliances of limited input such as laboratory burners and gas lights.
9. Recirculating direct-fired industrial air heaters as provided for in Section 612.

Where the appliances listed in Items 5 through 9 above are installed so that the aggregate input rating exceeds 20 British thermal units (Btu) per hour per cubic foot (207 watts per m³) of volume of the room or space in which such appliances are installed, one or more shall be provided with venting systems or other approved means for conveying the vent gases to the outdoor atmosphere so that the aggregate input rating of the remaining unvented appliances does not exceed 20 Btu per hour per cubic foot (207 watts per m³). Where the room or space in which the appliance is installed is directly connected to another room or space by a doorway, archway, or other opening of comparable size that cannot be closed, the volume of such adjacent room or space shall be permitted to be included in the calculations.

501.9 Chimney entrance. Connectors shall connect to a masonry chimney flue at a point not less than 12 inches (304.8 mm) or one chimney diameter above the lowest portion of the interior of the chimney flue.

501.10 Connections to exhauster. Appliance connections to a chimney or vent equipped with a power exhauster shall be made on the inlet side of the exhauster. Joints and vent systems on the positive pressure side of the exhauster shall be sealed to prevent flue-gas leakage as specified by the manufacturer's instructions for the exhauster or in accordance with this code.

501.11 Masonry chimneys. Masonry chimneys utilized to vent appliances shall be located, constructed and sized as specified in the manufacturer's instructions for the appliances being vented and Section 503.

501.12 Residential and low-heat appliances flue lining systems. Flue lining systems for use with residential-type and low-heat appliances shall be limited to the following:

1. Clay flue lining complying with the requirements of ASTM C 315 or equivalent. Clay flue lining shall be installed in accordance with the *New York City Building Code*.
2. Listed chimney lining systems complying with UL 1777 (new and existing chimneys) or ULC-S635 (existing chimneys) or ULC-S640 (new chimneys).
3. Other approved materials that will resist, without cracking, softening or corrosion, flue gases and condensate at temperatures up to 1,800°F (982.2°C).

501.13 Category I appliance flue lining systems. Flue lining systems for use with Category I appliances shall be limited to the following:

1. Flue lining systems complying with Section 501.12.
2. Chimney lining systems listed and labeled for use with gas appliances with draft hoods and other Category I gas appliances listed and labeled for use with Type B vents.

501.14 Category II, III and IV appliance venting systems. The design, sizing and installation of vents for Category II, III and IV appliances shall be in accordance with the appliance manufacturer's instructions.

501.15 Existing chimneys and vents. Where an appliance is permanently disconnected from an existing chimney or vent, or where an appliance is connected to an existing chimney or vent during the process of a new installation, the chimney or vent shall comply with Sections 501.15.1 through 501.15.4.

501.15.1 Size. The chimney or vent shall be resized as necessary to control flue gas condensation in the interior of the chimney or vent and to provide the appliance or appliances served with the required draft. For Category I appliances, the resizing shall be in accordance with Section 502.

501.15.2 Flue passageways. The flue gas passageway shall be free of obstructions and combustible deposits and shall be cleaned if previously used for venting a solid or liquid fuel-burning appliance or fireplace. The flue liner, chimney inner wall or vent inner wall shall be continuous and shall be free of cracks, gaps, perforations or other damage or deterioration that would allow the escape of combustion products, including gases, moisture and creosote.

501.15.3 Cleanout. Masonry chimney flues shall be provided with a cleanout opening having a minimum height of 6 inches (152.4 mm). The upper edge of the opening shall be located not less than 6 inches (152.4 mm) below the lowest chimney inlet opening. The cleanout shall be provided with a tight-fitting, noncombustible cover of a minimum size of 8 inches by 8 inches (203.2 mm by 203.2 mm).

501.15.4 Clearances. Chimneys and vents shall have airspace clearance to combustibles in accordance with the *New York City Building Code* and the chimney or vent manufacturer's instructions.

Exception: Masonry chimneys without the required air-space clearances shall be permitted to be used if lined or relined with a chimney lining system listed for use in chimneys with reduced clearances in accordance with UL 1777 or ULC-S635. The chimney clearance shall be not less than permitted by the terms of the chimney liner listing and the manufacturer's instructions.

501.15.4.1 Fireblocking. Noncombustible fireblocking shall be provided in accordance with the *New York City Building Code*.

501.15.5 Testing. Testing of existing chimneys shall be in accordance with Section 1705.32 of the *New York City Building Code*.

501.16 Drains. A drain shall be provided for all chimneys and gas vents to remove rain water and condensation. The drain shall be not less than 1 inch (25.4 mm) in diameter and shall be equipped with an appropriately-sized p-trap with automatic trap seal primer in accordance with Section 1002 of the *New York City Plumbing Code* or a float drain trap installed in accordance with the manufacturer's requirements. The drain shall be sized by the design engineer and shall be suitable for the chimney area. For listed chimneys and gas vents, the connection tap into the chimney shall be determined by the manufacturer and connected to the drain piping in accordance with the listing and instructions. On all outdoor chimneys and gas vents, the connection and drain shall be installed indoors as close as practicable to the chimney base to prevent freezing.

501.17 Outdoor chimneys and vents. Outdoor portions of chimneys or vents shall be provided with integral R-8 insulation or be provided with an R-8 insulation enclosure, where exposed to the outdoors for more than 5 feet (1524 mm).

501.17.1 Type B vents. Type B vents shall not be considered exposed to the outdoors in the following conditions:

1. A type B vent of a listed chimney system passing through an unused masonry chimney provided with a sealed cap that prevents airflow within the confines of the masonry chimney; or
2. A type B vent passing through an unventilated enclosure or chase insulated to R-8.

SECTION FGC 502 VENTS

502.1 General. Vents, except as provided in Section 503.7, shall be listed and labeled. Type B and BW vents shall be tested in accordance with UL 441 and labeled or field fabricated in accordance with NFPA 211. Type L vents shall be tested in accordance with UL 641. Vents for Category II, III and IV appliances shall be tested in accordance with UL 1738. Plastic piping shall be listed and installed in accordance with the terms of its listing and the appliance manufacturer's instructions.

502.2 Connectors required. Connectors shall be used to connect appliances to the vertical chimney or vent, except where the chimney or vent is attached directly to the appliance. Vent connector size, material, construction and installation shall be in accordance with Section 503.

502.3 Vent application. The application of vents shall be in accordance with Table 503.4.

502.4 Insulation shield. Where vents pass through insulated assemblies, an insulation shield constructed of steel having a minimum thickness of 0.0187 inch (0.4712 mm) (No. 26 gage) shall be installed to provide clearance between the vent and the insulation material. The clearance shall be not less than the clearance to combustibles specified by the vent manufacturer's instructions. Where vents pass through attic space, the shield shall terminate not less than 2 inches (50.8 mm) above the insulation materials and shall be secured in place to prevent displacement. Insulation shields provided as part of a listed vent system shall be installed in accordance with the manufacturer's instructions.

502.5 Installation. Vent systems shall be sized, installed and terminated in accordance with the vent and appliance manufacturer's instructions and Section 503.

502.6 Support of vents. All portions of vents shall be adequately supported for the design and weight of the materials employed.

502.7 Protection against physical damage. In concealed locations, where a vent is installed through holes or notches in studs, joists, rafters or similar members less than 1½ inches (38.1 mm) from the nearest edge of the member, the vent shall be protected by shield plates. Protective steel shield plates having a minimum thickness of 0.0575 inch (1.461 mm) (No. 16 gage) shall cover the area of the vent where the member is notched or bored and shall extend not less than 4 inches (101.6 mm) above sole plates, below top plates and to each side of a stud, joist or rafter.

502.7.1 Door swing. Appliance and equipment vent terminals shall be located such that doors cannot swing within 12 inches (304.8 mm) horizontally of the vent terminal. Door stops or closers shall not be installed to obtain this clearance. Means of protecting the clearance that are not easily overridden or removed may be used to obtain this clearance.

SECTION FGC 503 VENTING OF APPLIANCES

503.1 General. The venting of appliances shall be in accordance with Sections 503.2 through 503.16.

503.2 Venting systems required. Except as permitted in Sections 503.2.1 through 503.2.4 and 501.8, all appliances shall be connected to venting systems.

503.2.1 Ventilating hoods. Ventilating hoods and exhaust systems shall be permitted to be used to vent appliances installed in commercial applications in accordance with Section 503.3.4 and to vent industrial appliances, such as where the process itself requires fume disposal.

503.2.2 Reserved.

503.2.3 Direct-vent appliances. Listed direct-vent appliances shall be considered properly vented where installed in accordance with the terms of its listing, the manufacturer's instructions and Section 503.8.

503.2.4 Appliances with integral vents. An appliance incorporating integral venting means shall be considered properly vented where installed in accordance with its listing, the manufacturer's instructions and Section 503.8.

503.2.5 Incinerators. Commercial-industrial-type incinerators shall be vented in accordance with NFPA 82.

503.3 Design and construction. Venting systems shall be designed and constructed so as to convey all flue and vent gases to the outdoors.

503.3.1 Appliance draft requirements. A venting system shall satisfy the draft requirements of the appliance in accordance with the manufacturer’s instructions.

503.3.2 Design and construction. Appliances required to be vented shall be connected to a venting system designed and installed in accordance with the provisions of Sections 503.4 through 503.16.

503.3.3 Mechanical draft systems. Mechanical draft systems shall comply with the following:

- 1. Mechanical draft systems shall be listed and shall be installed in accordance with the terms of their listing and both the appliance and the mechanical draft system manufacturer’s instructions.
- 2. Appliances requiring venting shall be permitted to be vented by means of mechanical draft systems of either forced or induced draft design.
- 3. Forced draft systems and all portions of induced draft systems under positive pressure during operation shall be designed and installed so as to prevent leakage of flue or vent gases into a building.
- 4. Vent connectors serving appliances vented by natural draft shall not be connected to any portion of mechanical draft systems operating under positive pressure.
- 5. Where a mechanical draft system is employed, provisions shall be made to prevent the flow of gas to the main burners when the draft system is not performing so as to satisfy the operating requirements of the appliance for safe performance.
- 6. The exit terminals of mechanical draft systems shall be not less than 7 feet (2133.6 mm) above finished ground level where located adjacent to public walkways and shall be located as specified in Section 503.8, Items 1 and 2.

503.3.4 Ventilating hoods and exhaust systems. Ventilating hoods and exhaust systems shall be permitted to be used to vent appliances installed in commercial applications. Where automatically operated appliances, other than commercial cooking appliances, are vented through a ventilating hood or exhaust system equipped with a damper or with a power means of exhaust, provisions shall be made to allow the flow of gas to the main burners only when the damper is open to a position to properly vent the appliance and when the power means of exhaust is in operation.

503.3.5 Air ducts and furnace plenums. Venting systems shall not extend into or pass through any fabricated air duct or furnace plenum.

503.3.6 Above-ceiling air-handling spaces. Where a venting system passes through an above-ceiling air-handling space or other nonducted portion of an air-handling system, such space shall be accessible for inspection. The venting system shall conform to one of the following requirements:

- 1. The venting system shall be a listed special gas vent; other venting system serving a Category III or Category IV appliance; or other positive pressure vent, with joints sealed in accordance with the appliance or vent manufacturer’s instructions.
- 2. The venting system shall be installed such that fittings and joints between sections are not installed in the above-ceiling space.
- 3. The venting system shall be installed in a sealed metal conduit or enclosure with sealed joints separating the interior of the conduit or enclosure from the ceiling space.

503.4 Type of venting system to be used. The type of venting system to be used shall be in accordance with Table 503.4.

TABLE 503.4
TYPE OF VENTING SYSTEM TO BE USED

GAS UTILIZATION EQUIPMENT	TYPE OF VENTING SYSTEM
Listed Category I equipment Listed equipment equipped with draft hood Equipment listed for use with Type B gas vent	Type B gas vent (Section 503.6) Chimney (Section 503.5) Single-wall metal pipe (Section 503.7) Listed chimney lining system for gas venting (Section 503.5.3)
Listed vented wall furnaces	Type B-W gas vent (Sections 503.6, 608)
Category II equipment	As specified or furnished by manufacturers of listed equipment (Sections 503.4.1, 503.4.2)
Category III equipment	As specified or furnished by manufacturers of listed equipment (Sections 503.4.1, 503.4.2)
Category IV equipment	As specified or furnished by manufacturers of listed equipment (Sections 503.4.1, 503.4.2)
Incinerators	In accordance with NFPA 82
Equipment which may be converted to use of solid fuel	Chimney (Section 503.5)

Unlisted combination gas- and oil-burning equipment	Chimney (Section 503.5)
Listed combination gas- and oil-burning equipment	Type L vent (Section 503.6) or chimney (Section 503.5)
Combination gas- and solid fuel-burning equipment	Chimney (Section 503.5)
Equipment listed for use with chimneys only	Chimney (Section 503.5)
Unlisted equipment	Chimney (Section 503.5)
Decorative appliance in vented fireplace	Chimney
Direct vent equipment	See Section 503.2.3
Appliances with integral vent	See Section 503.2.4

503.4.1 Plastic piping. Plastic piping used for venting appliances listed for use with such venting materials shall be listed and installed in accordance with the terms of its listing and the manufacturers' instructions. Installation shall be in accordance with the *New York City Building Code*. PVC shall not be permitted.

503.4.1.1 Plastic vent joints. Plastic pipe and fittings used to vent appliances shall be installed in accordance with the appliance manufacturer's instructions. Where a primer is required, it shall be of a contrasting color.

503.4.2 Special gas vent. Special gas vent shall be listed and installed in accordance with the terms of the special gas vent listing and the manufacturers' instructions.

503.4.2.1 Inspection of special gas vents. Before replacing an existing appliance or connecting a vent connector to a special gas vent, the special gas vent shall be examined to ascertain that it is clear and free of obstructions.

503.4.2.2 Unsafe special gas vents. Where inspection reveals that an existing special gas vent is not safe for the intended application, it shall be repaired or replaced.

503.4.2.3 Test run. All special gas vents shall be test run by the registered design professional or special inspector responsible for the testing under operating conditions to demonstrate fire safety and the complete exhausting of smoke and the products of combustion to the outer air. The results of such test run shall be certified as correct by the registered design professional or special inspector responsible for the test and shall be submitted in writing to the department.

503.4.2.4 Requirement of a smoke test. A smoke test shall be conducted in accordance with Section 503.4.2.5. Any faults or leaks found shall be corrected. Such smoke test shall be witnessed by a representative of the commissioner. In the alternative, the commissioner may accept a test report instead of requiring that it be witnessed by a representative of the commissioner. Such test report of the registered design professional or special inspector responsible for the test which shall be submitted in writing to the department.

503.4.2.5 Smoke test. To determine the tightness of vent construction, a smoke test shall be conducted in accordance with the following conditions and requirements:

1. The equipment, materials, power and labor necessary for such test shall be furnished by, and at the expense of, either the owner or holder of the work permit.
2. If the test shows any evidence of leakage or other defects, such leakage or other defects shall be corrected in accordance with the requirements of this chapter, and the test shall be repeated until the results are satisfactory.
3. The vent shall be filled with a thick penetrating smoke produced by one or more smoke machines, or smoke bombs, or other equivalent method. As the smoke appears at the stack opening on the roof, such opening shall be tightly closed and a pressure equivalent to $\frac{1}{2}$ inch (12.7 mm) column of water measured at the base of the special gas vent shall be applied. The test shall be applied for a length of time sufficient to permit the inspection of the special gas vent.
4. Testing and inspection of existing and new negative pressure chimneys and vents shall be in accordance with Section 1705.32 of the *New York City Building Code*.

503.5 Masonry, metal, and factory-built chimneys. Masonry, metal and factory-built chimneys shall comply with Sections 503.5.1 through 503.5.10.

503.5.1 Factory-built chimneys. Factory-built chimneys shall be installed in accordance with their listing and the manufacturer's instructions. Factory-built chimneys used to vent appliances that operate at positive vent pressure shall be listed for such application.

503.5.2 Metal chimneys. Metal chimneys shall be built and installed in accordance with NFPA 211.

503.5.3 Masonry chimneys. Masonry chimneys shall be built and installed in accordance with Chapter 21 of the *New York City Building Code* and NFPA 211 and shall be lined with approved clay flue lining, a listed chimney lining system or other approved material that will resist corrosion, erosion, softening or cracking from vent gases at temperatures up to 1,800°F (982.2°C).

Exception: Masonry chimney flues serving listed gas appliances with draft hoods, Category I appliances and other gas appliances listed for use with Type B vents shall be permitted to be lined with a chimney lining system specifically listed for use only with such appliances. The liner shall be installed in accordance with the liner manufacturer's instructions and the terms of the listing. A permanent identifying label shall be attached at the point where the connection is to be made to the liner. The label shall read: "This chimney liner is for appliances that burn gas only. Do not connect to solid or liquid fuel-burning appliances or incinerators."

For installation of gas vents in existing masonry chimneys, see Section 503.6.3.

§§§503.5.4 **Chimney termination.** Chimneys serving gas-fired equipment shall comply with the appliance listing, the manufacturer's instructions, Figure 503.5.4 and the following requirements:

1. Chimneys, vents and flues serving appliances or fireplaces with outlet temperatures less than 600°F (315.6°C) shall extend not less than 3 feet (914.4 mm) above the highest construction, such as a roof ridge, parapet wall, or penthouse within 10 feet (3048 mm) of the chimney, vent or flue outlet, whether the construction is on the same building as the chimney or on another building. However, such constructions do not include other chimneys, vents, or open structural framing. Any chimneys, vents or flues located beyond 10 feet (3048 mm) from such construction, but not more than the distance determined by Equation 5-1, shall be at least as high as the construction.
2. Chimneys serving appliances or fireplaces with outlet temperatures between 600°F (315.6°C) and 1000°F (537.8°C) shall extend not less than 10 feet (3048 mm) above the highest construction, such as a roof ridge, or parapet wall or penthouse within 20 feet (6096 mm) of the chimney outlet, whether the construction is on the same building as the chimney or on another building. However, such constructions do not include other chimneys, and vents or open structural framing. Any chimneys located beyond 20 feet (6096 mm) from such construction, but not more than the distance determined by Equation 5-1, shall be at least as high as the construction.
3. Chimneys serving appliances or fireplaces with outlet temperatures greater than 1000°F (537.8°C) shall extend not less than 20 feet (6096 mm) above the highest construction, such as roof ridge, parapet wall, penthouse, or other obstruction within 50 feet (15 240 mm) of the chimney outlet, whether the construction is on the same building as the chimney or in another building. However, such constructions do not include other chimneys, vents, or open structural framing. Any chimneys located beyond 50 feet (15 240 mm) from such construction, but not more than the distance determined by Equation 5-1, shall be at least as high as the construction.
4. Termination caps shall not be permitted. A drain shall be installed in accordance with Section 801.22 of the *New York City Mechanical Code*. A positive means shall be provided to prevent water from entering the appliance.

Exception: Termination caps shall be permitted on listed factory-built chimneys unless otherwise prohibited by the *New York City Air Pollution Control Code*.

5. Decorative shrouds shall not be installed at the termination of factory-built chimneys except where such shrouds are listed and labeled for use with the specific factory-built chimney system and are installed in accordance with the manufacturers' instructions.
6. The following equation shall be used for determining the distances referred to in Items 1, 2 and 3 of this section.

$$D = F \times \sqrt{A} \quad \text{(Equation 5-1)}$$

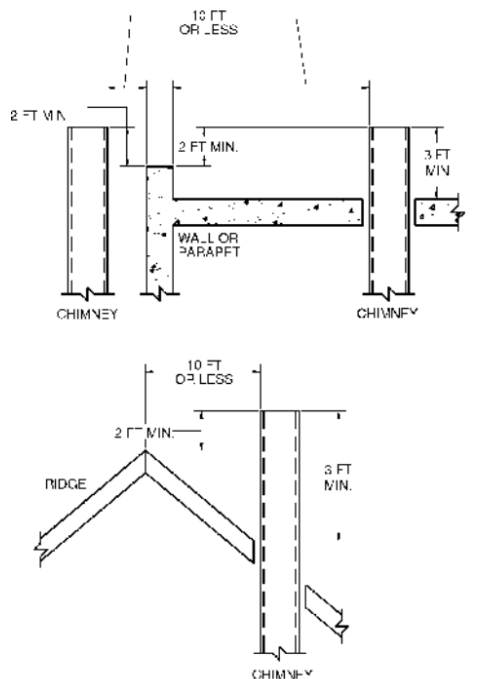
where:

D = Distance, in feet, measured from the center of the chimney, vent or flue outlet to the nearest edge of the construction. If a single chimney is divided into multiple smaller flues or chimneys, measure from the center of the chimney outlet that is closest to the nearest edge of the construction.

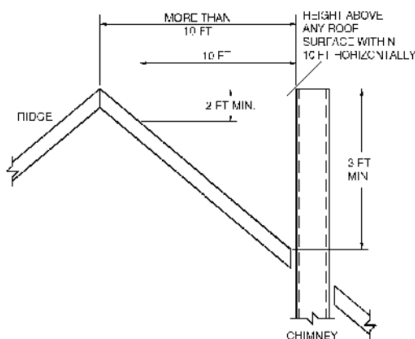
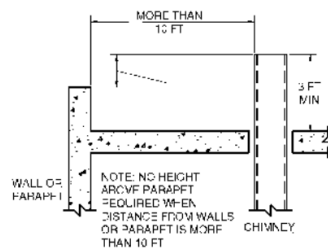
F = Value determined from table below.

A = Free area, in square inches, of chimney flue space outlet. If a single chimney is divided into multiple smaller flues or chimneys, the total aggregate free area of such flue and chimney outlets shall be used to calculate "A".

"F" FACTOR FOR DETERMINING CHIMNEY DISTANCES			
Type of Fuel	"F" Factor		
	600°F (315.6°C) and less	600°F (315.6°C) to 1000°F (537.8°C)	Greater than 1000°F (537.8°C)
Gas	2	2	3



A. TERMINATION 10 FT OR LESS FROM RIDGE, WALL, OR PARAPET



C. TERMINATION MORE THAN 10 FT FROM RIDGE, WALL, OR PARAPET
For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

FIGURE 503.5.4

TYPICAL TERMINATION LOCATIONS FOR CHIMNEYS AND SINGLE-WALL METAL PIPES SERVING RESIDENTIAL-TYPE AND LOW-HEAT APPLIANCES

503.5.5 Size of chimneys. The effective area of a chimney venting system serving listed appliances with draft hoods, Category I appliances, and other appliances listed for use with Type B vents shall be determined in accordance with one of the following methods:

1. The provisions of Section 504.
2. For sizing an individual chimney venting system for a single appliance with a draft hood, the effective areas of the vent connector and chimney flue shall be not less than the area of the appliance flue collar or draft hood outlet, nor greater than seven times the draft hood outlet area.

3. For sizing a chimney venting system connected to two appliances with draft hoods, the effective area of the chimney flue shall be not less than the area of the larger draft hood outlet plus 50 percent of the area of the smaller draft hood outlet, nor greater than seven times the smallest draft hood outlet area.
4. Chimney venting systems using mechanical draft shall be sized in accordance with approved engineering methods.
5. Other approved engineering methods.

503.5.6 Inspection of chimneys. Before replacing an existing appliance or connecting a vent connector to a chimney, the chimney passageway shall be examined to ascertain that it is clear and free of obstructions and it shall be cleaned if previously used for venting solid or liquid fuel-burning appliances or fireplaces.

503.5.6.1 Chimney lining. Chimneys shall be lined in accordance with NFPA 211.

Exception: Where an existing chimney complies with Sections 503.5.6 through 503.5.6.5 and its sizing is in accordance with Section 503.5.5, its continued use shall be allowed where the appliance vented by such chimney is replaced by an appliance of similar type, input rating and efficiency.

503.5.6.2 Cleanouts. Cleanouts shall be examined to determine if they will remain tightly closed when not in use.

503.5.6.3 Unsafe chimneys. Where inspection reveals that an existing chimney is not safe for the intended application, it shall be repaired, rebuilt, lined, relined or replaced with a vent or chimney to conform to NFPA 211 and it shall be suitable for the appliances to be vented.

503.5.6.4 Test run. All new chimneys shall be test run by the registered design professional responsible for the testing under operating conditions to demonstrate fire safety and the complete exhausting of smoke and the products of combustion to the outer air. The results of such test run shall be certified as correct by the registered design professional responsible for the test and shall be submitted in writing to the department.

Exception: A test run in accordance with this section may be conducted and certified to the department by the permit-holder when the work is performed as part of a limited alteration application, as defined in Section 28-101.5 of the Administrative Code. The test run shall not require a registered design professional or special inspector.

503.5.6.5 Requirement of a smoke test. A smoke test shall be made as outlined below. Any faults or leaks found shall be corrected. Such smoke test shall be witnessed by a representative of the commissioner. In lieu thereof, the commissioner may accept the test report of the registered design professional or special inspector responsible for the test that shall be submitted in writing to the department.

Exception: A smoke test may be performed by or under the supervision of a permit-holder when the work is performed as part of a limited alteration application, as defined in Section 28-101.5 of the Administrative Code. Such test shall not be required to be witnessed by the department, registered design professional, or special inspector.

503.5.6.5.1 Smoke test. To determine the tightness of chimney construction, a smoke test shall be made in accordance with the following conditions and requirements:

1. The equipment, materials, power and labor necessary for such test shall be furnished by, and at the expense of, the owner or holder of the work permit.
2. If the test shows any evidence of leakage or other defects, such defects shall be corrected in accordance with the requirement of this chapter and the test shall be repeated until the results are satisfactory.
3. Method of test. The chimney shall be filled with a thick penetrating smoke produced by one or more smoke machines, or smoke bombs, or other equivalent method. As the smoke appears at the stack opening on the roof, such opening shall be tightly closed and a pressure equivalent to $\frac{1}{2}$ inch (12.7 mm) column of water measured at the base of the stack, shall be applied. The test shall be applied for a length of time sufficient to permit the inspection of the chimney.

503.5.7 Chimneys serving appliances burning other fuels. Chimneys serving appliances burning other fuels shall comply with Sections 503.5.7.1 through 503.5.7.4.

503.5.7.1 Solid fuel-burning appliances. An appliance shall not be connected to a chimney flue serving a separate appliance designed to burn solid fuel.

503.5.7.2 Liquid fuel-burning appliances. Where one chimney flue serves gas appliances and liquid fuel-burning appliances, the appliances shall be connected through separate openings or shall be connected through a single opening where joined by a suitable fitting located as close as practical to the chimney. Where two or more openings are provided into one chimney flue, they shall be at different levels. Where the appliances are automatically controlled, they shall be equipped with safety shutoff devices.

503.5.7.3 Combination gas- and solid fuel-burning appliances. A combination gas- and solid fuel-burning appliance shall be permitted to be connected to a single chimney flue where equipped with a manual reset device to shut off gas to the main burner in the event of sustained back draft or flue gas spillage. The chimney flue shall be sized to properly vent the appliance.

503.5.7.4 Combination gas- and oil fuel-burning appliances. A listed combination gas- and oil fuel-burning appliance shall be permitted to be connected to a single chimney flue. The chimney flue shall be sized to properly vent the appliance.

503.5.8 Support of chimneys. All portions of chimneys shall be supported for the design and weight of the materials employed. Factory-built chimneys shall be supported and spaced in accordance with their listings and the manufacturer's instructions.

503.5.9 Cleanouts. Where a chimney that formerly carried flue products from liquid or solid fuel-burning appliances is used with an appliance using fuel gas, an accessible cleanout shall be provided. The cleanout shall have a tight-fitting cover and shall be installed at the base of all chimneys in accordance with the rules of the *New York City Department of Environmental Protection*. The cleanout shall be gas tight if the chimney is to be of positive pressure.

503.5.10 Space surrounding lining or vent. The remaining space surrounding a chimney liner, gas vent, special gas vent or plastic piping installed within a masonry chimney flue shall not be used to vent another appliance. The insertion of another liner or vent within the chimney as provided in this code and the liner or vent manufacturer's instructions shall not be prohibited.

The remaining space surrounding a chimney liner, gas vent, special gas vent or plastic piping installed within a masonry, metal or factory-built chimney, shall not be used to supply combustion air. Such space shall not be prohibited from supplying combustion air to direct-vent appliances designed for installation in a solid fuel-burning fireplace and installed in accordance with the listing and the manufacturer's instructions.

503.6 Gas vents. Gas vents shall comply with Sections 503.6.1 through 503.6.13 (see Section 202, Definitions).

503.6.1 Installation, general. Gas vents shall be installed in accordance with the terms of their listings and manufacturer's instructions.

503.6.2 Type B-W vent capacity. A Type B-W gas vent shall have a listed capacity not less than that of the listed vented wall furnace to which it is connected.

503.6.3 Gas vents installed within masonry chimneys. Gas vents installed within masonry chimneys shall be installed in accordance with the terms of their listing and the manufacturer's instructions. Gas vents installed within masonry chimneys shall be identified with a permanent label installed at the point where the vent enters the chimney. The label shall contain the following language: "This gas vent is for appliances that burn gas. Do not connect to solid or liquid fuel-burning appliances or incinerators."

503.6.4 Gas vent terminations. A gas vent shall terminate in accordance with one of the following:

1. Gas vents that are 12 inches (304.8 mm) or less in size and located not less than 8 feet (2438.4 mm) from a vertical wall or similar obstruction shall terminate above the roof in accordance with Figure 503.6.4.
2. Gas vents that are over 12 inches (304.8 mm) in size or are located less than 8 feet (2438.4 mm) from a vertical wall or similar obstruction shall terminate not less than 2 feet (609.6 mm) above the highest point where they pass through the roof and not less than 2 feet (609.6 mm) above any portion of a building within 10 feet (3048 mm) horizontally.
3. As provided for direct-vent systems in Section 503.2.3.
4. As provided for appliances with integral vents in Section 503.2.4.
5. As provided for mechanical draft systems in Section 503.3.3.
6. As provided for ventilating hoods and exhaust systems in Section 503.3.4.

However, such constructions do not include chimneys or other vents, or open structural framing. The vent shall be as high as such construction which is located beyond 10 feet (3048 mm) from the vent and up to and including the distance determined by the following formula:

$$D = 2\sqrt{A} \quad \text{(Equation 5-2)}$$

where:

D = Distance, in feet, measured from the center of the vent outlet to the nearest edge of the construction.

A = Free area, in square inches, of vent flue space.

503.6.4.1 Decorative shrouds. Decorative shrouds shall not be installed at the termination of gas vents except where such shrouds are listed for use with the specific gas venting system and are installed in accordance with the manufacturer's instructions.

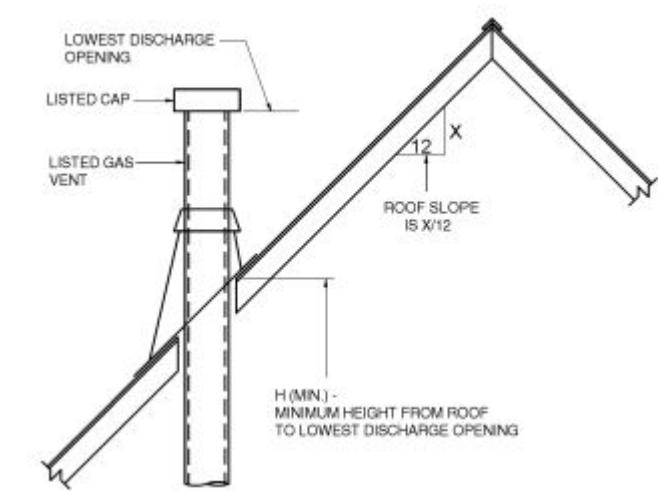
All gas vents shall terminate in an approved vent or cowl, which shall prevent downdrafts and prevent rain and debris from entering the vent.

503.6.5 Minimum height. A Type B or L gas vent shall terminate not less than 5 feet (1524 mm) in vertical height above the highest connected appliance draft hood or flue collar. A Type B-W gas vent shall terminate not less than 12 feet (3657.6 mm) in vertical height above the bottom of the wall furnace.

503.6.6 Roof terminations. Gas vents shall extend through the roof flashing, roof jack or roof thimble and terminate with a listed cap or listed roof assembly.

503.6.7 Forced air inlets. Gas vents shall terminate not less than 3 feet (914.4 mm) above any forced air inlet located within 10 feet (3048 mm).

503.6.8 Exterior wall penetrations. A gas vent extending through an exterior wall shall not terminate adjacent to the wall or below eaves or parapets, except as provided in Sections 503.2.4 and 503.3.3.



ROOF PITCH	H (min) ft
Flat to 6/12	1.0
Over 6/12 to 7/12	1.25
Over 7/12 to 8/12	1.5
Over 8/12 to 9/12	2.0
Over 9/12 to 10/12	2.5
Over 10/12 to 11/12	3.25
Over 11/12 to 12/12	4.0
Over 12/12 to 14/12	5.0
Over 14/12 to 16/12	6.0
Over 16/12 to 18/12	7.0
Over 18/12 to 20/12	7.5
Over 20/12 to 21/12	8.0

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

FIGURE 503.6.4
TERMINATION LOCATIONS FOR GAS VENTS WITH
LISTED CAPS 12 INCHES OR LESS IN SIZE AT
LEAST 8 FEET FROM A VERTICAL WALL

503.6.9 Size of gas vents. Venting systems shall be sized and constructed in accordance with Section 504 or other approved engineering methods acceptable to the commissioner. Gas vents shall also comply with the manufacturers’ instructions.

503.6.9.1 Category I appliances. The sizing of natural draft venting systems serving one or more listed appliances equipped with a draft hood or appliances listed for use with Type B gas vent, installed in a single story of a building, shall be in accordance with one of the following methods:

1. The provisions of Section 504.
2. For sizing an individual gas vent for a single, draft-hood-equipped appliance, the effective area of the vent connector and the gas vent shall be not less than the area of the appliance draft hood outlet, nor greater than seven times the draft hood outlet area.

3. For sizing a gas vent connected to two appliances with draft hoods, the effective area of the vent shall be not less than the area of the larger draft hood outlet plus 50 percent of the area of the smaller draft hood outlet, nor greater than seven times the smaller draft hood outlet area.

4. Engineering practices acceptable to the commissioner.

503.6.9.2 Vent offsets. Type B and L vents sized in accordance with Item 2 or 3 of Section 503.6.9.1 shall extend in a generally vertical direction with offsets not exceeding 45 degrees (0.79 rad), except that a vent system having not more than one 60-degree (1.04 rad) offset shall be permitted. Any angle greater than 45 degrees (0.79 rad) from the vertical is considered horizontal. The total horizontal distance of a vent plus the horizontal vent connector serving draft-hood-equipped appliances shall be not greater than 75 percent of the vertical height of the vent.

503.6.9.3 Category II, III, and IV appliances. The sizing of gas vents or plastic pipe specified by the appliance manufacturer for Category II, III, and IV appliances shall be in accordance with the appliance manufacturer's instructions.

503.6.9.4 Mechanical draft. Chimney venting systems using mechanical draft shall be designed and sized to develop a positive flow adequate to carry all products of combustion to the outside atmosphere.

503.6.10 Multistory prohibited. Common venting systems for appliances located on more than one floor level shall be prohibited.

503.6.11 Support of gas vents. Gas vents shall be supported and spaced in accordance with their listings and the manufacturer's instructions.

503.6.12 Marking. Gas vents shall be permanently identified by a label attached to the wall or ceiling at a point where the vent connector enters the gas vent. The label shall read:

"This gas vent is for appliances that burn gas. Do not connect to solid or liquid fuel-burning appliances or incinerators."

503.6.13 Fastener penetrations. Screws, rivets and other fasteners shall not penetrate the inner wall of double-wall gas vents, except at the transition from an appliance draft hood outlet, a flue collar or a single-wall metal connector to a double-wall vent.

503.7 Single-wall metal pipe. Single-wall metal pipe vents shall comply with Sections 503.7.1 through 503.7.13.

503.7.1 Construction. Single-wall metal pipe shall be constructed of galvanized sheet steel not less than 0.0304 inch (0.77 mm) thick, or other approved, noncombustible, corrosion-resistant material.

503.7.2 Climate protection. Uninsulated single-wall metal pipe shall not be used outdoors for venting appliances.

503.7.3 Termination. Single-wall metal pipe shall terminate at least 5 feet (1524 mm) in vertical height above the highest connected appliance draft hood outlet or flue collar. Single-wall metal pipe shall extend not less than 2 feet (609.6 mm) above the highest point where it passes through a roof of a building and not less than 2 feet (609.6 mm) higher than any portion of a building within a horizontal distance of 10 feet (3048 mm) (see Figure 503.5.4). An approved cap or roof assembly shall be attached to the terminus of a single-wall metal pipe (see also Section 503.7.9, Item 3). However, such constructions do not include chimneys or other vents, or open structural framing. The single-wall metal pipe shall be as high as such construction which is located beyond 10 feet (3048 mm) from the vent and up to and including the distance determined by Equation 5-2.

503.7.4 Limitations of use. Single-wall metal pipe shall be used only for runs directly from the space in which the appliance is located through the roof or exterior wall to the outdoor atmosphere.

503.7.5 Roof penetrations. A pipe passing through a roof shall extend without interruption through the roof flashing, roof jack, or roof thimble. Where a single-wall metal pipe passes through a roof constructed of combustible material, a noncombustible, non-ventilating thimble shall be used at the point of passage. The thimble shall extend not less than 18 inches (457.2 mm) above and 6 inches (152.4 mm) below the roof with the annular space open at the bottom and closed only at the top. The thimble shall be sized in accordance with Section 503.7.7.

503.7.6 Installation. Single-wall metal pipe shall not originate in any unoccupied attic or concealed space and shall not pass through any attic, inside wall, concealed space, or floor. The installation of a single-wall metal pipe through an exterior combustible wall shall comply with Section 503.7.7.

503.7.7 Single-wall penetrations of combustible walls. A single-wall metal pipe shall not pass through a combustible exterior wall unless guarded at the point of passage by a ventilated metal thimble not smaller than the following:

1. For listed appliances with draft hoods and appliances listed for use with Type B gas vents, the thimble shall be not less than 4 inches (101.6 mm) larger in diameter than the metal pipe. Where there is a run of not less than 6 feet (1828.8 mm) of metal pipe in the open between the draft hood outlet and the thimble, the thimble shall be permitted to be not less than 2 inches (50.8 mm) larger in diameter than the metal pipe.
2. For unlisted appliances having draft hoods, the thimble shall be not less than 6 inches (152.4 mm) larger in diameter than the metal pipe.

3. For residential and low-heat appliances, the thimble shall be not less than 12 inches (304.8 mm) larger in diameter than the metal pipe.

Exception: In lieu of thimble protection, all combustible material in the wall shall be removed a sufficient distance from the metal pipe to provide the specified clearance from such metal pipe to combustible material. Any material used to close up such opening shall be noncombustible.

503.7.8 Clearances. Minimum clearances from single-wall metal pipe to combustible material shall be in accordance with Table 503.10.5. The clearance from single-wall metal pipe to combustible material shall be permitted to be reduced where the combustible material is protected as specified for vent connectors in Table 308.2.

503.7.9 Size of single-wall metal pipe. A venting system constructed of single-wall metal pipe shall be sized in accordance with one of the following methods and the appliance manufacturer's instructions:

1. For a draft-hood-equipped appliance, in accordance with Section 504.
2. For a venting system for a single appliance with a draft hood, the areas of the connector and the pipe each shall be not less than the area of the appliance flue collar or draft hood outlet, whichever is smaller. The vent area shall be not greater than seven times the draft hood outlet area.
3. Other approved engineering methods.

503.7.10 Pipe geometry. Any shaped single-wall metal pipe shall be permitted to be used, provided that its equivalent effective area is equal to the effective area of the round pipe for which it is substituted, and provided that the minimum internal dimension of the pipe is not less than 2 inches (50.8 mm).

503.7.11 Termination capacity. The vent cap or a roof assembly shall have a venting capacity of not less than that of the pipe to which it is attached.

503.7.12 Support of single-wall metal pipe. All portions of single-wall metal pipe shall be supported for the design and weight of the material employed.

503.7.13 Marking. Single-wall metal pipe shall comply with the marking provisions of Section 503.6.12.

†††**503.8 Venting system termination location.** The location of venting system terminations shall comply with the following:

1. Gas venting systems shall be extended at least 3 feet (914.4 mm) above the highest construction, such as a roof ridge, parapet wall, or penthouse, within 10 feet (3048 mm) of the vent outlet whether the construction is on the same building as the chimney or on another building. However, such constructions do not include chimneys or other vents, or open structural framing. The vent shall be as high as such construction which is located beyond 10 feet (3048 mm) from the vent and up to and including the distance determined by Equation 5-2.

Exception: Horizontally terminated direct-vent appliances and integral vent appliances approved by the commissioner and installed in accordance with the manufacturer's instructions and Section 503.8, Item 3.

2. Where permitted, through-the-wall vents for Category I, II, III and IV appliances and noncategorized condensing appliances shall not terminate over public walkways or over an area where condensate or vapor could create a nuisance or hazard or could be detrimental to the operation of regulators, relief valves or other equipment.
3. Horizontal terminations shall only be allowed if they are in a nonhazardous location and if the appliance has a sealed combustion chamber (direct vent) or integral vent in accordance with the appliance listing and manufacturer's instructions. In addition, horizontal terminations shall comply with the following requirements:
 - 3.1. Where located adjacent to walkways, the termination shall be not less than 7 feet (2133.6 mm) above the level of the walkway.
 - 3.2. Vents shall terminate at least 3 feet (914.4 mm) above any forced air inlet, other than the forced air inlet for the subject direct vent or integral vent appliance, located within 10 feet (3048 mm).
 - 3.3. The vent system shall terminate at least 4 feet (1219.2 mm) below, 4 feet (1219.2 mm) horizontally from or 1 foot (304.8 mm) above any door, window or gravity air inlet into the building.
 - 3.4. The vent termination point shall not be located closer than 3 feet (914.4 mm) to an interior corner formed by two walls perpendicular to each other.
 - 3.5. The vent termination shall not be mounted directly above or within 3 feet (914.4 mm) horizontally from any gas or electric metering, regulating, venting relief equipment or other building opening.
 - 3.6. The bottom of the vent termination shall be located at least 36 inches (914.4 mm) above finished grade.
 - 3.7. The maximum heat input of an appliance served by single horizontal vent termination shall be 350,000 Btu/h (102.6 kW), unless otherwise approved by the commissioner.
 - 3.8. The maximum heat input of all appliances served by horizontal vent terminations located within a 10 feet (3048 mm) radius shall be 350,000 Btu/h (102.6 kW), unless otherwise approved by the commissioner.

3.9. The vent termination shall be located a minimum of 4 feet (1219.2 mm) from the lot line or from adjacent buildings. The termination shall be installed in accordance with the gas vent manufacturer's listing and instructions.

503.9 Condensation drainage. Provisions shall be made to collect and dispose of condensate from venting systems serving Category II and IV appliances and noncategorized condensing appliances in accordance with Section 307.

503.10 Vent connectors for Category I appliances. Vent connectors for Category I appliances shall comply with Sections 503.10.1 through 503.10.14.

503.10.1 Where required. A vent connector shall be used to connect an appliance to a gas vent, chimney, or single-wall metal pipe, except where the gas vent, chimney, or single-wall metal pipe is directly connected to the appliance.

503.10.2 Materials. Vent connectors shall be constructed in accordance with Sections 503.10.2.1 through 503.10.2.3[‡].

503.10.2.1 General. A vent connector shall be made of noncombustible corrosion-resistant material capable of withstanding the vent gas temperature produced by the appliance and of sufficient thickness to withstand physical damage.

503.10.2.2 Vent connectors located in unconditioned areas. Where the vent connector used for an appliance having a draft hood or a Category I appliance is located in or passes through attics, crawl spaces or other unconditioned spaces, that portion of the vent connector shall be listed Type B, Type L or listed vent material having equivalent insulation properties.

***503.10.2.3 Vent connector installation.** Vent connectors shall be installed in accordance with one of the following:

1. Connectors with flue gas temperatures less than 500°F (260°C) that pass through walls or partitions of combustible construction shall be installed in accordance with the terms of their listing, the manufacturer's instructions and Section 803.10.4 of the New York City Mechanical Code.
2. Connectors of appliances with flue gas temperatures less than 500°F (260°C) that pass through walls or partitions of combustible construction shall be insulated with a field-applied wrap assembly in accordance with Section 803.10.4.2 of the New York City Mechanical Code.
3. Residential-type appliance connectors installed in accordance with Section 503.10.2.3.1.
4. Low-heat appliance connectors installed in accordance with Section 503.10.2.3.2.
5. Medium-heat appliance connectors installed in accordance with Section 503.10.2.3.3.

*Section 503.10.2.3 was added by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

‡‡‡503.10.2.3.1 Residential-type appliance connectors. Where vent connectors for residential-type appliances are not installed in attics or other unconditioned spaces, connectors for listed appliances having draft hoods, appliances having draft hoods and equipped with listed conversion burners and Category I appliances shall be one of the following:

1. Type B or Type L vent material.
2. Galvanized sheet steel not less than 0.018 inch (0.46 mm) thick.
3. Aluminum (1100 or 3003 alloy or equivalent) sheet not less than 0.027 inch (0.69 mm) thick.
4. Stainless steel sheet not less than 0.012 inch (0.31 mm) thick.
5. Smooth interior wall metal pipe having resistance to heat and corrosion equal to or greater than that of Item 2, 3 or 4.
6. A listed vent connector.

Vent connectors shall not be covered with insulation.

‡‡‡503.10.2.3.2 Low-heat appliances. A vent connector for low-heat appliances shall be a factory-built chimney section or steel pipe having resistance to heat and corrosion equivalent to that for the appropriate galvanized pipe as specified in Table 503.10.2.3.2[‡]. Factory-built chimney sections shall be joined together in accordance with the chimney manufacturer's instructions.

TABLE 503.10.2.3.2[‡]
MINIMUM THICKNESS FOR GALVANIZED STEEL VENT
CONNECTORS FOR LOW-HEAT APPLIANCES

DIAMETER OF CONNECTOR (inches)	MINIMUM THICKNESS (inch)
Less than 6	0.019
6 to less than 10	0.023
10 to 12 inclusive	0.029
14 to 16 inclusive	0.034
Over 16	0.056

For SI: 1 inch = 25.4 mm.

†††**503.10.2.3.3 Medium-heat appliances.** Vent connectors for medium-heat appliances shall be constructed of factory-built medium-heat chimney sections or steel of a thickness not less than that specified in Table 503.10.2.3.3[‡] and shall comply with the following:

1. A steel vent connector for an appliance with a vent gas temperature in excess of 1000°F (537.8°C) measured at the entrance to the connector shall be lined with medium-duty fire brick (ASTM C 64, Type F), or the equivalent.
2. The lining shall be not less than 2½ inches (63.5 mm) thick for a vent connector having a diameter or greatest cross-sectional dimension of 18 inches (457.2 mm) or less.
3. The lining shall be not less than 4½ inches (114.3 mm) thick laid on the 4½-inch (114.3 mm) bed for a vent connector having a diameter or greatest cross-sectional dimension greater than 18 inches (457.2 mm).
4. Factory-built chimney sections, if employed, shall be joined together in accordance with the chimney manufacturer's instructions.

TABLE 503.10.2.3.3[‡]
MINIMUM THICKNESS FOR STEEL VENT
CONNECTORS FOR MEDIUM-HEAT APPLIANCES

VENT CONNECTOR SIZE		MINIMUM THICKNESS (inch)
DIAMETER (inches)	AREA (square inches)	
Up to 14	Up to 154	0.053
Over 14 to 16	154 to 201	0.067
Over 16 to 18	201 to 254	0.093
Over 18	Larger than 254	0.123

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm².

503.10.3 Size of vent connector. Vent connectors shall be sized in accordance with Sections 503.10.3.1 through 503.10.3.5.

503.10.3.1 Single draft hood and fan-assisted. A vent connector for an appliance with a single draft hood or for a Category I fan-assisted combustion system appliance shall be sized and installed in accordance with Section 504 or other approved engineering methods.

503.10.3.2 Multiple draft hoods. For a single appliance having more than one draft hood outlet or flue collar, the manifold shall be constructed according to the instructions of the appliance manufacturer. Where there are no instructions, the manifold shall be designed and constructed in accordance with approved engineering practices. As an alternate method, the effective area of the manifold shall equal the combined area of the flue collars or draft hood outlets and the vent connectors shall have a minimum 1-foot (304.8 mm) rise.

503.10.3.3 Multiple appliances. Where two or more appliances are connected to a common vent or chimney, each vent connector shall be sized in accordance with Section 504 or other approved engineering methods.

As an alternative method applicable only when all of the appliances are draft hood equipped, each vent connector shall have an effective area not less than the area of the draft hood outlet of the appliance to which it is connected.

503.10.3.4 Common connector/manifold. Where two or more appliances are vented through a common vent connector or vent manifold, the common vent connector or vent manifold shall be located at the highest level consistent with available headroom and the required clearance to combustible materials and shall be sized in accordance with Section 504 or other approved engineering methods.

As an alternate method applicable only where there are two draft hood-equipped appliances, the effective area of the common vent connector or vent manifold and all junction fittings shall be not less than the area of the larger vent connector plus 50 percent of the area of the smaller flue collar outlet.

503.10.3.5 Size increase. Where the size of a vent connector is increased to overcome installation limitations and obtain connector capacity equal to the appliance input, the size increase shall be made at the appliance draft hood outlet.

503.10.4 Two or more appliances connected to a single vent or chimney. Where two or more vent connectors enter a common gas vent, chimney flue or single-wall metal pipe, the smaller connector shall enter at the highest level consistent with the available headroom or clearance to combustible material. Vent connectors serving Category I appliances shall not be connected to any portion of a mechanical draft system operating under positive static pressure, such as those serving Category III or IV appliances.

503.10.4.1 Two or more openings. Where two or more openings are provided into one chimney flue or vent, the openings shall be at different levels, or the connectors shall be attached to the vertical portion of the chimney or vent at an angle of 45 degrees (0.79 rad) or less relative to the vertical in accordance with Figure 503.10.4.1.

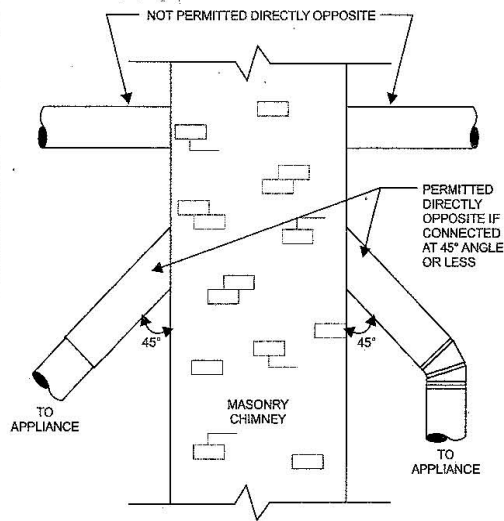


FIGURE 503.10.4.1
OPPOSING OPENINGS IN CHIMNEY

503.10.5 Clearance. Minimum clearances from vent connectors to combustible material shall be in accordance with Table 503.10.5.

Exception: The clearance between a vent connector and combustible material shall be permitted to be reduced where the combustible material is protected as specified for vent connectors in Table 308.2.

TABLE 503.10.5^a
CLEARANCES FOR CONNECTORS

APPLIANCE	MINIMUM DISTANCE FROM COMBUSTIBLE MATERIAL			
	Listed Type B gas vent material	Listed Type L vent material	Single-wall metal pipe	Factory-built chimney sections
Listed appliances with draft hoods and appliances listed for use with Type B gas vents	As listed	As listed	6 inches	As listed
Residential boilers and furnaces with listed gas conversion burner and with draft hood	6 inches	6 inches	9 inches	As listed
Residential appliances listed for use with Type L vents	Not permitted	As listed	9 inches	As listed
Listed gas-fired toilets	Not permitted	As listed	As listed	As listed
Unlisted residential appliances with draft hood	Not permitted	6 inches	9 inches	As listed
Residential and low-heat appliances other than above	Not permitted	9 inches	18 inches	As listed
Medium-heat appliances	Not permitted	Not permitted	36 inches	As listed

For SI: 1 inch = 25.4 mm.

a. These clearances shall apply unless the manufacturer's installation instructions for a listed appliance or connector specify different clearances, in which case the listed clearances shall apply.

503.10.6 Joints. Joints between sections of connector and connections to flue collars and draft hood outlets shall be tight-fitting and fastened by one of the following methods

1. Galvanized steel – Overlapping joint with sheet metal screws or rivets at a minimum of 4 locations for diameters not greater than 12 inches (304.8mm) and 8 locations for diameters greater than 12 inches (304.8 mm).
2. Carbon steel – Overlapping joint with $\frac{1}{2}$ inch (12.7 mm) tack welds at a minimum of 4 locations for diameters up to 12 inches (304.8 mm) and 8 locations for diameters larger than 12 inches (304.8 mm).
3. Carbon steel – continuous butt welded perimeter.
4. Vent connectors of listed vent material assembled and connected to flue collars or draft hood outlets in accordance with the manufacturer's instructions.

503.10.7 Slope. A vent connector shall be installed without dips or sags and shall slope upward toward the vent or chimney not less than $\frac{1}{4}$ inch per foot (21 mm/m).

Exception: Vent connectors attached to a mechanical draft system installed in accordance with the manufacturers' instructions.

503.10.8 Length of vent connector. The maximum horizontal length of a single-wall connector shall be 75 percent of the height of the chimney or vent except for engineered systems. The maximum horizontal length of a Type B double-wall connector shall be 100 percent of the height of the chimney or vent except for engineered systems.

503.10.9 Support. A vent connector shall be supported in accordance with the manufacturer's recommendations for the design and weight of the material employed to maintain clearances and prevent physical damage and separation of joints.

503.10.10 Chimney connection. Where entering a flue in a masonry or metal chimney, the vent connector shall be installed above the extreme bottom to avoid stoppage. Where a thimble or slip joint is used to facilitate removal of the connector, the connector shall be firmly attached to or inserted into the thimble or slip joint to prevent the connector from falling out. Means shall be employed to prevent the connector from entering so far as to restrict the space between its end and the opposite wall of the chimney flue (see Section 501.9).

503.10.11 Inspection. The entire length of a vent connector shall be provided with ready access for inspection, cleaning and replacement.

503.10.12 Fireplaces. A vent connector shall not be connected to a chimney flue serving a fireplace.

503.10.13 Passage through ceilings, floors or walls. Single-wall metal pipe connectors shall not pass through any wall, floor or ceiling except as permitted by Section 503.7.4.

503.10.14 Medium-heat connectors. Vent connectors for medium-heat appliances shall not pass through walls or partitions constructed of combustible material.

503.11 Vent connectors for Category II, III, and IV appliances. Vent connectors for Category II, III, and IV appliances shall be as specified for the venting systems in accordance with Section 503.4.

503.12 Draft hoods and draft controls. The installation of draft hoods and draft controls shall comply with Sections 503.12.1 through 503.12.7.

503.12.1 Appliances requiring draft hoods. Vented appliances shall be installed with draft hoods.

Exception: Direct-vent appliances, fan-assisted combustion system appliances, appliances requiring chimney draft for operation, appliances equipped with blast, power, or pressure burners that are not listed for use with draft hoods, and appliances designed for forced venting.

503.12.2 Installation. A draft hood supplied with or forming a part of a listed vented appliance shall be installed without alteration, exactly as furnished and specified by the appliance manufacturer.

503.12.2.1 Draft hood required. If a draft hood is not supplied by the appliance manufacturer where one is required, a draft hood shall be installed, shall be of a listed or approved type and, in the absence of other instructions, shall be of the same size as the appliance flue collar. Where a draft hood is required with a conversion burner, it shall be of a listed or approved type.

503.12.2.2 Special design draft hood. Where it is determined that a draft hood of special design is needed or preferable for a particular installation, the installation shall be in accordance with the recommendations of the appliance manufacturer and shall be approved by the department.

503.12.3 Draft control devices. Where a draft control device is part of the appliance or is supplied by the appliance manufacturer, it shall be installed in accordance with the manufacturer's instructions. In the absence of manufacturer's instructions, the device shall be attached to the flue collar of the appliance or as near to the appliance as practical.

503.12.4 Additional devices. Appliances requiring a controlled chimney draft shall be permitted to be equipped with a listed double-acting barometric-draft regulator installed and adjusted in accordance with the manufacturers' instructions.

503.12.5 Location. Draft hoods and barometric draft regulators shall be installed in the same room or enclosure as the appliance in such a manner as to prevent any difference in pressure between the hood or regulator and the combustion air supply.

503.12.6 Positioning. Draft hoods and draft regulators shall be installed in the position for which they were designed with reference to the horizontal and vertical planes and shall be located so that the relief opening is not obstructed by any part of the appliance or adjacent construction. The appliance and its draft hood shall be located so that the relief opening is accessible for checking vent operation.

503.12.7 Clearance. A draft hood shall be located so its relief opening is not less than 6 inches (152.4 mm) from any surface except that of the appliance it serves and the venting system to which the draft hood is connected. Where a greater or lesser clearance is indicated on the appliance label, the clearance shall be not less than that specified on the label. Such clearances shall not be reduced.

503.12.8 Thermal safety (spill) switches. Thermal safety (spill) switches shall be installed on barometric-draft regulators and all other appurtenances that allow dilution air into chimneys or vents. Thermal safety (spill) switches shall be interlocked with all of the appliances connected to the same chimney or gas vent.

503.13 Manually operated dampers. A manually operated damper shall not be placed in the vent connector for any appliance. Fixed baffles shall not be classified as manually operated dampers.

503.14 Automatically operated vent dampers. An automatically operated vent damper shall be of a listed type complete with safety controls to prove open position of the damper and/or appropriate draft before appliance operation. The automatically operated dampers shall be installed in accordance with the manufacturer's instructions. An automatic vent damper device shall not be installed on an existing appliance unless the appliance is listed and labeled and the device is installed in accordance with the terms of its listing. The name of the installer and date of installation shall be marked on a label affixed to the damper device.

503.15 Obstructions. Devices that retard the flow of vent gases shall not be installed in a vent connector, chimney or vent. The following shall not be considered as obstructions:

1. Draft regulators and safety controls specifically listed for installation in venting systems and installed in accordance with the terms of their listing and manufacturer's instructions.
2. Approved draft regulators and safety controls that are designed and installed in accordance with approved engineering methods.
3. Listed heat reclaimers and automatically operated vent dampers installed in accordance with the terms of their listing and manufacturer's instructions.
4. Approved economizers, heat reclaimers and recuperators installed in venting systems of appliances not required to be equipped with draft hoods, provided that the appliance manufacturer's instructions cover the installation of such a device in the venting system and performance in accordance with Sections 503.3 and 503.3.1 is obtained.
5. Vent dampers serving listed appliances installed in accordance with Sections 504.2.1 and 504.3.1 or other approved engineering methods.

503.16 Outside wall penetrations. Where vents, including those for direct-vent appliances, penetrate outside walls of buildings, the annular spaces around such penetrations shall be permanently sealed using approved materials to prevent entry of combustion products into the building.

SECTION FGC 504 SIZING OF CATEGORY I APPLIANCE VENTING SYSTEMS

504.1 Definitions. The following definitions apply to the tables in this section.

APPLIANCE CATEGORIZED VENT DIAMETER/AREA. The minimum vent area/diameter permissible for Category I appliances to maintain a non-positive vent static pressure when tested in accordance with nationally recognized standards.

FAN-ASSISTED COMBUSTION SYSTEM. An appliance equipped with an integral mechanical means to either draw or force products of combustion through the combustion chamber or heat exchanger.

FAN Min. The minimum input rating of a Category I fan-assisted appliance attached to a vent or connector.

FAN Max. The maximum input rating of a Category I fan-assisted appliance attached to a vent or connector.

NAT Max. The maximum input rating of a Category I draft-hood-equipped appliance attached to a vent or connector.

FAN + FAN. The maximum combined appliance input rating of two or more Category I fan-assisted appliances attached to the common vent.

FAN + NAT. The maximum combined appliance input rating of one or more Category I fan-assisted appliances and one or more Category I draft-hood-equipped appliances attached to the common vent.

NA. Vent configuration is not allowed due to potential for condensate formation or pressurization of the venting system, or not applicable due to physical or geometric restraints.

NAT + NAT. The maximum combined appliance input rating of two or more Category I draft-hood-equipped appliances attached to the common vent.

504.2 Application of single-appliance vent Tables 504.2(1) through 504.2(6). The application of Tables 504.2(1) through 504.2(6) shall be subject to the requirements of Sections 504.2.1 through 504.2.17.

504.2.1 Vent obstructions. These venting tables shall not be used where obstructions, as described in Section 503.15, are installed in the venting system. The installation of vents serving listed appliances with vent dampers shall be in accordance with the appliance manufacturer's instructions or in accordance with the following:

1. The maximum capacity of the vent system shall be determined using the "NAT Max" column.

2. The minimum capacity shall be determined as if the appliance were a fan-assisted appliance, using the “FAN Min” column to determine the minimum capacity of the vent system. Where the corresponding “FAN Min” is “NA,” the vent configuration shall not be permitted and an alternative venting configuration shall be utilized.

504.2.2 Minimum size. Where the vent size determined from the tables is smaller than the appliance draft hood outlet or flue collar, the smaller size shall be permitted to be used provided that all of the following requirements are met:

1. The total vent height (H) is not less than 10 feet (3048 mm).
2. Vents for appliance draft hood outlets or flue collars 12 inches (304.8 mm) in diameter or smaller are not reduced more than one table size.
3. Vents for appliance draft hood outlets or flue collars larger than 12 inches (304.8 mm) in diameter are not reduced more than two table sizes.
4. The maximum capacity listed in the tables for a fan-assisted appliance is reduced by 10 percent ($0.90 \times$ maximum table capacity).
5. The draft hood outlet is greater than 4 inches (101.6 mm) in diameter. Do not connect a 3-inch-diameter (76.2 mm) vent to a 4-inch-diameter (101.6 mm) draft hood outlet. This provision shall not apply to fan-assisted appliances.

TABLE 504.2(1)
TYPE B DOUBLE-WALL GAS VENT

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Connected directly to vent

HEIGHT (H) (feet)	LATERAL (L) (feet)	VENT DIAMETER—(D) inches																					
		3			4			5			6			7			8			9			
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																					
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	
6	0	0	78	46	0	152	86	0	251	141	0	375	205	0	524	285	0	698	370	0	897	470	
	2	13	51	36	18	97	67	27	157	105	32	232	157	44	321	217	53	425	285	63	543	370	
	4	21	49	34	30	94	64	39	153	103	50	227	153	66	316	211	79	419	279	93	536	362	
	6	25	46	32	36	91	61	47	149	100	59	223	149	78	310	205	93	413	273	110	530	354	
8	0	0	84	50	0	165	94	0	276	155	0	415	235	0	583	320	0	780	415	0	1,006	537	
	2	12	57	40	16	109	75	25	178	120	28	263	180	42	365	247	50	483	322	60	619	418	
	5	23	53	38	32	103	71	42	171	115	53	255	173	70	356	237	83	473	313	99	607	407	
	8	28	49	35	39	98	66	51	164	109	64	247	165	84	347	227	99	463	303	117	596	396	
10	0	0	88	53	0	175	100	0	295	166	0	447	255	0	631	345	0	847	450	0	1,096	585	
	2	12	61	42	17	118	81	23	194	129	26	289	195	40	402	273	48	533	355	57	684	457	
	5	23	57	40	32	113	77	41	187	124	52	280	188	68	392	263	81	522	346	95	671	446	
	10	30	51	36	41	104	70	54	176	115	67	267	175	88	376	245	104	504	330	122	651	427	
15	0	0	94	58	0	191	112	0	327	187	0	502	285	0	716	390	0	970	525	0	1,263	682	
	2	11	69	48	15	136	93	20	226	150	22	339	225	38	475	316	45	633	414	53	815	544	
	5	22	65	45	30	130	87	39	219	142	49	330	217	64	463	300	76	620	403	90	800	529	
	10	29	59	41	40	121	82	51	206	135	64	315	208	84	445	288	99	600	386	116	777	507	
	15	35	53	37	48	112	76	61	195	128	76	301	198	98	429	275	115	580	373	134	755	491	
20	0	0	97	61	0	202	119	0	349	202	0	540	307	0	776	430	0	1,057	575	0	1,384	752	
	2	10	75	51	14	149	100	18	250	166	20	377	249	33	531	346	41	711	470	50	917	612	
	5	21	71	48	29	143	96	38	242	160	47	367	241	62	519	337	73	697	460	86	902	599	
	10	28	64	44	38	133	89	50	229	150	62	351	228	81	499	321	95	675	443	112	877	576	
	15	34	58	40	46	124	84	59	217	142	73	337	217	94	481	308	111	654	427	129	853	557	
	20	48	52	35	55	116	78	69	206	134	84	322	206	107	464	295	125	634	410	145	830	537	

(continued)

TABLE 504.2(1)—continued
TYPE B DOUBLE-WALL GAS VENT

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Connected directly to vent

HEIGHT (H) (feet)	LATERAL (L) (feet)	VENT DIAMETER—(D) inches																				
		3			4			5			6			7			8			9		
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																				
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max
30	0	0	100	64	0	213	128	0	374	220	0	587	336	0	853	475	0	1,173	650	0	1,548	855

	2	9	81	56	13	166	112	14	283	185	18	432	280	27	613	394	33	826	535	42	1,072	700
	5	21	77	54	28	160	108	36	275	176	45	421	273	58	600	385	69	811	524	82	1,055	688
	10	27	70	50	37	150	102	48	262	171	59	405	261	77	580	371	91	788	507	107	1,028	668
	15	33	64	NA	44	141	96	57	249	163	70	389	249	90	560	357	105	765	490	124	1,002	648
	20	56	58	NA	53	132	90	66	237	154	80	374	237	102	542	343	119	743	473	139	977	628
	30	NA	NA	NA	73	113	NA	88	214	NA	104	346	219	131	507	321	149	702	444	171	929	594
50	0	0	101	67	0	216	134	0	397	232	0	633	363	0	932	518	0	1,297	708	0	1,730	952
	2	8	86	61	11	183	122	14	320	206	15	497	314	22	715	445	26	975	615	33	1,276	813
	5	20	82	NA	27	177	119	35	312	200	43	487	308	55	702	438	65	960	605	77	1,259	798
	10	26	76	NA	35	168	114	45	299	190	56	471	298	73	681	426	86	935	589	101	1,230	773
	15	59	70	NA	42	158	NA	54	287	180	66	455	288	85	662	413	100	911	572	117	1,203	747
	20	NA	NA	NA	50	149	NA	63	275	169	76	440	278	97	642	401	113	888	556	131	1,176	722
	30	NA	NA	NA	69	131	NA	84	250	NA	99	410	259	123	605	376	141	844	522	161	1,125	670
100	0	NA	NA	NA	0	218	NA	0	407	NA	0	665	400	0	997	560	0	1,411	770	0	1,908	1,040
	2	NA	NA	NA	10	194	NA	12	354	NA	13	566	375	18	831	510	21	1,155	700	25	1,536	935
	5	NA	NA	NA	26	189	NA	33	347	NA	40	557	369	52	820	504	60	1,141	692	71	1,519	926
	10	NA	NA	NA	33	182	NA	43	335	NA	53	542	361	68	801	493	80	1,118	679	94	1,492	910
	15	NA	NA	NA	40	174	NA	50	321	NA	62	528	353	80	782	482	93	1,095	666	109	1,465	895
	20	NA	NA	NA	47	166	NA	59	311	NA	71	513	344	90	763	471	105	1,073	653	122	1,438	880
	30	NA	NA	NA	NA	NA	NA	78	290	NA	92	483	NA	115	726	449	131	1,029	627	149	1,387	849
	50	NA	NA	NA	NA	NA	NA	NA	NA	NA	147	428	NA	180	651	405	197	944	575	217	1,288	787

(continued)

**TABLE 504.2(1)—continued
TYPE B DOUBLE-WALL GAS VENT**

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Connected directly to vent

HEIGHT (H) (feet)	LATERAL (L) (feet)	VENT DIAMETER—(D) inches																															
		10				12				14				16				18				20				22				24			
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																															
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT					
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max					
6	0	0	1,121	570	0	1,645	850	0	2,267	1,170	0	2,983	1,530	0	3,802	1,960	0	4,721	2,430	0	5,737	2,950	0	6,853	3,520								
	2	75	675	455	103	982	650	138	1,346	890	178	1,769	1,170	225	2,250	1,480	296	2,782	1,850	360	3,377	2,220	426	4,030	2,670								
	4	110	668	445	147	975	640	191	1,338	880	242	1,761	1,160	300	2,242	1,475	390	2,774	1,835	469	3,370	2,215	555	4,023	2,660								
	6	128	661	435	171	967	630	219	1,330	870	276	1,753	1,150	341	2,235	1,470	437	2,767	1,820	523	3,363	2,210	618	4,017	2,650								
8	0	0	1,261	660	0	1,858	970	0	2,571	1,320	0	3,399	1,740	0	4,333	2,220	0	5,387	2,750	0	6,555	3,360	0	7,838	4,010								

	2	71	770	515	98	1,12 4	745	130	1,543	1,02 0	168	2,03 0	1,34 0	212	2,58 4	1,70 0	278	3,19 6	2,11 0	336	3,88 2	2,56 0	401	4,63 4	3,05 0
	5	115	758	503	154	1,11 0	733	199	1,528	1,01 0	251	2,01 3	1,33 0	311	2,56 3	1,68 5	398	3,18 0	2,09 0	476	3,86 3	2,54 5	562	4,61 2	3,04 0
	8	137	746	490	180	1,09 7	720	231	1,514	1,00 0	289	2,00 0	1,32 0	354	2,55 2	1,67 0	450	3,16 3	2,07 0	537	3,85 0	2,53 0	630	4,60 2	3,03 0
10	0	0	1,37 7	720	0	2,03 6	1,06 0	0	2,825	1,45 0	0	3,74 2	1,92 5	0	4,78 2	2,45 0	0	5,95 5	3,05 0	0	7,25 4	3,71 0	0	8,68 2	4,45 0
	2	68	852	560	93	1,24 4	850	124	1,713	1,13 0	161	2,25 6	1,48 0	202	2,86 8	1,89 0	264	3,55 6	2,34 0	319	4,32 2	2,84 0	378	5,15 3	3,39 0
	5	112	839	547	149	1,22 9	829	192	1,696	1,10 5	243	2,23 8	1,46 1	300	2,84 9	1,87 1	382	3,53 6	2,31 8	458	4,30 1	2,81 8	540	5,13 2	3,37 1
	10	142	817	525	187	1,20 4	795	238	1,669	1,08 0	298	2,20 9	1,43 0	364	2,81 8	1,84 0	459	3,50 4	2,28 0	546	4,26 8	2,78 0	641	5,09 9	3,34 0
15	0	0	1,59 6	840	0	2,38 0	1,24 0	0	3,323	1,72 0	0	4,42 3	2,27 0	0	5,67 8	2,90 0	0	7,09 9	3,62 0	0	8,66 5	4,41 0	0	10,3 93	5,30 0
	2	63	1,01 9	675	86	1,49 5	985	114	2,062	1,35 0	147	2,71 9	1,77 0	186	3,46 7	2,26 0	239	4,30 4	2,80 0	290	5,23 2	3,41 0	346	6,25 1	4,08 0
	5	105	1,00 3	660	140	1,47 6	967	182	2,041	1,32 7	229	2,69 6	1,74 8	283	3,44 2	2,23 5	355	4,27 8	2,77 7	426	5,20 4	3,38 5	501	6,22 2	4,05 7
	10	135	977	635	177	1,44 6	936	227	2,009	1,28 9	283	2,65 9	1,71 2	346	3,40 2	2,19 3	432	4,23 4	2,73 9	510	5,15 9	3,34 3	599	6,17 5	4,01 9
	15	155	953	610	202	1,41 8	905	257	1,976	1,25 0	318	2,62 3	1,67 5	385	3,36 3	2,15 0	479	4,19 2	2,70 0	564	5,11 5	3,30 0	665	6,12 9	3,98 0
20	0	0	1,75 6	930	0	2,63 7	1,35 0	0	3,701	1,90 0	0	4,94 8	2,52 0	0	6,37 6	3,25 0	0	7,98 8	4,06 0	0	9,78 5	4,98 0	0	11,7 53	6,00 0
	2	59	1,15 0	755	81	1,69 4	1,10 0	107	2,343	1,52 0	139	3,09 7	2,00 0	175	3,95 5	2,57 0	220	4,91 6	3,20 0	269	5,98 3	3,91 0	321	7,15 4	4,70 0
	5	101	1,13 3	738	135	1,67 4	1,07 9	174	2,320	1,49 8	219	3,07 1	1,97 8	270	3,92 6	2,54 4	337	4,88 5	3,17 4	403	5,95 0	3,88 0	475	7,11 9	4,66 2
	10	130	1,10 5	710	172	1,64 1	1,04 5	220	2,282	1,46 0	273	3,02 9	1,94 0	334	3,88 0	2,50 0	413	4,83 5	3,13 0	489	5,89 6	3,83 0	573	7,06 3	4,60 0
	15	150	1,07 8	688	195	1,60 9	1,01 8	248	2,245	1,42 5	306	2,98 8	1,91 0	372	3,83 5	2,46 5	459	4,78 6	3,09 0	541	5,84 4	3,79 5	631	7,00 7	4,57 5
	20	167	1,05 2	665	217	1,57 8	990	273	2,210	1,39 0	335	2,94 8	1,88 0	404	3,79 1	2,43 0	495	4,73 7	3,05 0	585	5,79 2	3,76 0	689	6,95 3	4,55 0

(continued)

TABLE 504.2(2)
TYPE B DOUBLE-WALL GAS VENT

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Single-wall metal connector

HEIGHT T (H) (feet)	LATERAL AL (L) (feet)	VENT DIAMETER—(D) Inches																																			
		3				4				5				6				7				8				9				10				12			
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																																			
		FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT						
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max						
6	0	38	77	45	59	15 1	85	85	24 9	14 0	12 6	37 3	20 4	16 5	52 2	28 4	21 1	695	36 9	26 7	894	469	37 1	1,1 18	569	53 7	1.6 39	849									
	2	39	51	36	60	96	66	85	15 6	10 4	12 3	23 1	15 6	15 9	32 0	21 3	20 1	423	28 4	25 1	541	368	34 7	673	453	49 8	979	648									

		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max
30	0	0	1,97 7	1,06 0	0	3,00 4	1,55 0	0	4,25 2	2,17 0	0	5,72 5	2,92 0	0	7,42 0	3,77 0	0	9,34 1	4,75 0	0	11,4 83	5,85 0	0	13,84 8	7,06 0
	2	54	1,35 1	865	74	2,00 4	1,31 0	98	2,78 6	1,80 0	12 7	3,69 6	2,38 0	15 9	4,73 4	3,05 0	19 9	5,90 0	3,81 0	24 1	7,19 4	4,65 0	28 5	8,617	5,60 0
	5	96	1,33 2	851	12 7	1,98 1	1,28 9	16 4	2,75 9	1,77 5	20 6	3,66 6	2,35 0	25 2	4,70 1	3,02 0	31 2	5,86 3	3,78 3	37 3	7,15 5	4,62 2	43 9	8,574	5,55 2
	10	12 5	1,30 1	829	16 4	1,94 4	1,25 4	20 9	2,71 6	1,73 3	25 9	3,61 7	2,30 0	31 6	4,64 7	2,97 0	38 6	5,80 3	3,73 9	45 6	7,09 0	4,57 4	53 5	8,505	5,47 1
	15	14 3	1,27 2	807	18 7	1,90 8	1,22 0	23 7	2,67 4	1,69 2	29 2	3,57 0	2,25 0	35 4	4,59 4	2,92 0	43 1	5,74 4	3,69 5	50 7	7,02 6	4,52 7	59 0	8,437	5,39 1
	20	16 0	1,24 3	784	20 7	1,87 3	1,18 5	26 0	2,63 3	1,65 0	31 9	3,52 3	2,20 0	38 4	4,54 2	2,87 0	46 7	5,68 6	3,65 0	54 8	6,96 4	4,48 0	63 9	8,370	5,31 0
	30	19 5	1,18 9	745	24 6	1,80 7	1,13 0	30 5	2,55 5	1,58 5	36 9	3,43 3	2,13 0	44 0	4,44 2	2,78 5	54 0	5,57 4	3,56 5	63 5	6,84 2	4,37 5	73 9	8,239	5,22 5
50	0	0	2,23 1	1,19 5	0	3,44 1	1,82 5	0	4,93 4	2,55 0	0	6,71 1	3,44 0	0	8,77 4	4,46 0	0	11,1 29	5,63 5	0	13,7 67	6,94 0	0	16,69 4	8,43 0
	2	41	1,62 0	1,01 0	66	2,43 1	1,51 3	86	3,40 9	2,12 5	11 3	4,55 4	2,84 0	14 1	5,86 4	3,67 0	17 1	7,33 9	4,63 0	20 9	8,98 0	5,69 5	25 1	10,78 8	6,86 0
	5	90	1,60 0	996	11 8	2,40 6	1,49 5	15 1	3,38 0	2,10 2	19 1	4,52 0	2,81 3	23 4	5,82 6	3,63 9	28 3	7,29 5	4,59 7	33 6	8,93 3	5,65 4	39 4	10,73 7	6,81 8
	10	11 8	1,56 7	972	15 4	2,36 6	1,46 6	19 6	3,33 2	2,06 4	24 3	4,46 4	2,76 7	29 5	5,76 3	3,58 5	35 5	7,22 4	4,54 2	41 9	8,85 5	5,58 5	49 1	10,65 2	6,74 9
	15	13 6	1,53 6	948	17 7	2,32 7	1,43 7	22 2	3,28 5	2,02 6	27 4	4,40 9	2,72 1	33 0	5,70 1	3,53 4	39 6	7,15 5	4,51 1	46 5	8,77 9	5,54 6	54 2	10,57 0	6,71 0
	20	15 1	1,50 5	924	19 5	2,28 8	1,40 8	24 4	3,23 9	1,98 7	30 0	4,35 6	2,67 5	36 1	5,64 1	3,48 1	43 3	7,08 6	4,47 9	50 6	8,70 4	5,50 6	58 6	10,48 8	6,67 0
	30	18 3	1,44 6	876	23 2	2,21 4	1,34 9	28 7	3,15 0	1,91 0	34 7	4,25 3	2,63 1	41 2	5,52 3	3,43 1	49 4	6,95 3	4,42 1	57 7	8,55 7	5,44 4	67 2	10,32 8	6,60 3
100	0	0	2,49 1	1,31 0	0	3,92 5	2,05 0	0	5,72 9	2,95 0	0	7,91 4	4,05 0	0	10,4 85	5,30 0	0	13,4 54	6,70 0	0	16,8 17	8,60 0	0	20,57 8	10,3 00
	2	30	1,97 5	1,17 0	44	3,02 7	1,82 0	72	4,31 3	2,55 0	95	5,83 4	3,50 0	12 0	7,59 1	4,60 0	13 8	9,57 7	5,80 0	16 9	11,8 03	7,20 0	20 4	14,26 4	8,80 0
	5	82	1,95 5	1,15 9	10 7	3,00 2	1,80 3	13 6	4,28 2	2,53 1	17 2	5,79 7	3,47 5	20 8	7,54 8	4,56 6	24 5	9,52 8	5,76 9	29 3	11,7 48	7,16 2	34 1	14,20 4	8,75 6
	10	10 8	1,92 3	1,14 2	14 2	2,96 1	1,77 5	18 0	4,23 1	2,50 0	22 3	5,73 7	3,43 4	26 8	7,47 8	4,50 9	31 8	9,44 7	5,71 7	37 4	11,6 58	7,10 0	43 6	14,10 5	8,68 3
	15	12 6	1,89 2	1,12 4	16 3	2,92 0	1,74 7	20 6	4,18 2	2,46 9	25 2	5,67 8	3,39 2	30 4	7,40 9	4,45 1	35 8	9,36 7	5,66 5	41 8	11,5 69	7,03 7	48 7	14,00 7	8,61 0
	20	14 1	1,86 1	1,10 7	18 1	2,88 0	1,71 9	22 6	4,13 3	2,43 8	27 7	5,61 9	3,35 1	33 0	7,34 1	4,39 4	38 7	9,28 9	5,61 3	45 2	11,4 82	6,97 5	52 3	13,91 0	8,53 7
	30	17 0	1,80 2	1,07 1	21 5	2,80 3	1,66 3	26 5	4,03 7	2,37 5	31 9	5,50 5	3,26 7	37 8	7,20 9	4,27 9	44 6	9,13 6	5,50 9	51 4	11,3 10	6,85 0	59 2	13,72 0	8,39 1
	50	24 1	1,68 8	1,00 0	29 2	2,65 7	1,55 0	35 0	3,85 6	2,25 0	41 5	5,28 9	3,10 0	48 6	6,95 6	4,05 0	57 2	8,84 1	5,30 0	65 9	10,9 79	6,60 0	75 2	13,35 4	8,10 0

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

**TABLE 504.2(3)
MASONRY CHIMNEY**

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

HEIGHT	LATERAL	TYPE B DOUBLE-WALL CONNECTOR DIAMETER—(D) Inches to be used with chimney areas within the size limits at bottom									
		3	4	5	6	7	8	9	10	12	

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

HEIGHT (H) (feet)	LATERAL (L) (feet)	TYPE B DOUBLE-WALL CONNECTOR DIAMETER —(D) Inches to be used with chimney areas within the size limits at bottom								
		3	4	5	6	7	8	9	10	12
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H								

		FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NAT	FAN		NAT	
		Min	Ma x	Ma x	Min	Ma x	Ma x	Min	Ma x	Ma x	Min	Ma x	Ma x	Min	Ma x	Ma x	Min	Ma x	Ma x	Min	Max	Ma x	Min	Max	Max	Min	Max	Max	
30	2	NA	NA	41	NA	NA	82	NA	NA	13 7	NA	NA	21 6	47	58 1	30 3	57	76 2	42 1	68	985	55 8	81	1,2 40	717	111	1,7 93	1,1 12	
	5	NA	NA	NA	NA	NA	76	NA	NA	12 8	NA	NA	19 8	75	56 1	28 1	90	74 1	39 3	10 6	962	52 6	12 5	1,2 16	683	169	1,7 66	1,0 94	
	10	NA	NA	NA	NA	NA	67	NA	NA	11 5	NA	NA	18 4	NA	NA	26 3	11 5	70 9	37 3	13 5	927	50 0	15 8	1,1 76	648	210	1,7 21	1,0 25	
	15	NA	NA	NA	NA	NA	NA	NA	NA	10 7	NA	NA	17 1	NA	NA	24 3	NA	NA	35 3	15 6	893	47 6	18 1	1,1 39	621	239	1,6 79	981	
	20	NA	NA	NA	NA	NA	NA	NA	NA	91	NA	NA	15 9	NA	NA	22 7	NA	NA	33 2	17 6	860	45 0	20 3	1,1 03	592	264	1,6 38	940	
	30	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	18 8	NA	NA	28 8	NA	NA	41 6	24 9	1,0 35	555	318	1,5 60	877	
50	2	NA	NA	NA	NA	NA	92	NA	NA	16 1	NA	NA	25 1	NA	NA	35 1	51	84 0	47 7	61	1,1 06	63 3	72	1,4 13	812	99	2,0 80	1,2 43	
	5	NA	NA	NA	NA	NA	NA	NA	NA	15 1	NA	NA	23 0	NA	NA	32 3	83	81 9	44 5	98	1,0 83	59 6	11 6	1,3 87	774	155	2,0 52	1,2 25	
	10	NA	NA	NA	NA	NA	NA	NA	NA	13 8	NA	NA	21 5	NA	NA	30 4	NA	NA	42 4	12 6	1,0 47	56 7	14 7	1,3 47	733	195	2,0 06	1,1 47	
	15	NA	NA	NA	NA	NA	NA	NA	NA	12 7	NA	NA	19 9	NA	NA	28 2	NA	NA	40 0	14 6	1,0 10	53 9	17 0	1,3 07	702	222	1,9 61	1,0 99	
	20	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	18 5	NA	NA	26 4	NA	NA	37 6	16 5	977	51 1	19 0	1,2 69	669	246	1,9 16	1,0 50	
	30	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	32 7	NA	NA	46 8	23 3	1,1 96	623	295	1,8 32	984	
Minimum Internal Area of Chimney (square inches)		12			19			28			38			50			63			78			95			132			
Maximum Internal Area of Chimney (square inches)		Seven times the listed appliance categorized vent area, flue collar area or draft hood outlet area.																											

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

TABLE 504.2(4)
MASONRY CHIMNEY

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Single-wall metal connector

HEIGHT T (H) (feet)	LATERAL L (L) (feet)	SINGLE-WALL METAL CONNECTOR DIAMETER—(D) Inches to be used with chimney areas within the size limits at bottom																										
		3		4		5		6		7		8		9		10		12										
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																										
		FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NAT	FAN		NAT			
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max			
6	2	NA	NA	28	NA	NA	52	NA	NA	86	NA	NA	130	NA	NA	180	NA	NA	247	NA	NA	319	NA	NA	400	NA	NA	580
	5	NA	NA	25	NA	NA	48	NA	NA	81	NA	NA	116	NA	NA	164	NA	NA	230	NA	NA	297	NA	NA	375	NA	NA	560
8	2	NA	NA	29	NA	NA	55	NA	NA	93	NA	NA	145	NA	NA	197	NA	NA	265	NA	NA	349	382	725	445	549	1,021	650

	5	NA	NA	26	NA	NA	51	NA	NA	87	NA	NA	133	NA	NA	182	NA	NA	246	NA	NA	327	NA	NA	422	673	1,003	638
	8	NA	NA	23	NA	NA	47	NA	NA	82	NA	NA	126	NA	NA	174	NA	NA	237	NA	NA	317	NA	NA	408	747	985	621
10	2	NA	NA	31	NA	NA	61	NA	NA	102	NA	NA	161	NA	NA	220	216	518	297	271	654	387	373	808	490	536	1,142	722
	5	NA	NA	28	NA	NA	56	NA	NA	95	NA	NA	147	NA	NA	203	NA	NA	276	334	635	364	459	789	465	657	1,121	710
	10	NA	NA	24	NA	NA	49	NA	NA	86	NA	NA	137	NA	NA	189	NA	NA	261	NA	NA	345	547	758	441	771	1,088	665
15	2	NA	NA	35	NA	NA	67	NA	NA	113	NA	NA	178	166	473	249	211	611	335	264	776	440	362	965	560	520	1,373	840
	5	NA	NA	32	NA	NA	61	NA	NA	106	NA	NA	163	NA	NA	230	261	591	312	325	775	414	444	942	531	637	1,348	825
	10	NA	NA	27	NA	NA	54	NA	NA	96	NA	NA	151	NA	NA	214	NA	NA	294	392	722	392	531	907	504	749	1,309	774
	15	NA	NA	NA	NA	NA	46	NA	NA	87	NA	NA	138	NA	NA	198	NA	NA	278	452	692	372	606	873	481	841	1,272	738
20	2	NA	NA	38	NA	NA	73	NA	NA	123	NA	NA	200	163	520	273	206	675	374	258	864	490	252	1,079	625	508	1,544	950
	5	NA	NA	35	NA	NA	67	NA	NA	115	NA	NA	183	80	NA	252	255	655	348	317	842	461	433	1,055	594	623	1,518	930
	10	NA	NA	NA	NA	NA	59	NA	NA	105	NA	NA	170	NA	NA	235	312	622	330	382	806	437	517	1,016	562	733	1,475	875
	15	NA	NA	NA	NA	NA	NA	NA	NA	95	NA	NA	156	NA	NA	217	NA	NA	311	442	773	414	591	979	539	823	1,434	835
	20	NA	NA	NA	NA	NA	NA	NA	NA	80	NA	NA	144	NA	NA	202	NA	NA	292	NA	NA	392	663	944	510	911	1,394	800

(continued)

**TABLE 504.2(4)—continued
MASONRY CHIMNEY**

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Single-wall metal connector

HEIGHT T (H) (feet)	LATERAL AL (L) (feet)	SINGLE-WALL METAL CONNECTOR DIAMETER—(D) Inches to be used with chimney areas within the size limits at bottom																										
		3		4		5		6		7		8		9		10		12										
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																										
		FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NAT	FAN		NAT			
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max			
30	2	NA	NA	41	NA	NA	81	NA	NA	136	NA	NA	215	158	578	302	200	759	420	249	982	556	340	1,237	715	489	1,789	1,110
	5	NA	NA	NA	NA	NA	75	NA	NA	127	NA	NA	196	NA	NA	279	245	737	391	306	958	524	417	1,210	680	600	1,760	1,090
	10	NA	NA	NA	NA	NA	66	NA	NA	113	NA	NA	182	NA	NA	260	300	703	370	370	920	496	500	1,168	644	708	1,713	1,020
	15	NA	NA	NA	NA	NA	NA	NA	NA	105	NA	NA	168	NA	NA	240	NA	NA	349	428	884	471	572	1,128	615	798	1,668	975
	20	NA	NA	NA	NA	NA	NA	NA	NA	88	NA	NA	155	NA	NA	223	NA	NA	327	NA	NA	445	643	1,089	585	883	1,624	932
	30	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	182	NA	NA	281	NA	NA	408	NA	NA	544	1,055	1,539	865
50	2	NA	NA	NA	NA	NA	91	NA	NA	160	NA	NA	250	NA	NA	350	191	837	475	238	1,103	631	323	1,408	810	463	2,076	1,240
	5	NA	NA	NA	NA	NA	NA	NA	NA	149	NA	NA	228	NA	NA	321	NA	NA	442	293	1,078	593	398	1,381	770	571	2,044	1,220

	10	NA	NA	NA	NA	NA	NA	NA	NA	136	NA	NA	212	NA	NA	301	NA	NA	420	355	1,038	562	447	1,337	728	674	1,994	1,140	
	15	NA	NA	NA	NA	NA	NA	NA	NA	124	NA	NA	195	NA	NA	278	NA	NA	395	NA	NA	533	546	1,294	695	761	1,945	1,090	
	20	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	180	NA	NA	258	NA	NA	370	NA	NA	504	616	1,251	660	844	1,898	1,040	
	30	NA	NA	NA	NA	NA	48	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	318	NA	NA	458	NA	NA	610	1,009	1,805	970	
Minimum Internal Area of Chimney (square inches)		12			19			28			38			50			63			78			95			132			
Maximum Internal Area of Chimney (square inches)		Seven times the listed appliance categorized vent area, flue collar area or draft hood outlet area.																											

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

VENT CONNECTOR CAPACITY

VENT HEIGHT (H) (feet)	TYPE B DOUBLE-WALL COMMON VENT DIAMETER (D)—inches						
	4	5	6	7	8	9	10
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H						

	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT
6	92	81	65	140	116	103	204	161	147	309	248	200	404	314	260	547	434	335	672	520	410
8	101	90	73	155	129	114	224	178	163	339	275	223	444	348	290	602	480	378	740	577	465
10	110	97	79	169	141	124	243	194	178	367	299	242	477	377	315	649	522	405	800	627	495
15	125	112	91	195	164	144	283	228	206	427	352	280	556	444	365	753	612	465	924	733	565
20	136	123	102	215	183	160	314	255	229	475	394	310	621	499	405	842	688	523	1,035	826	640
30	152	138	118	244	210	185	361	297	266	547	459	360	720	585	470	979	808	605	1,209	975	740
50	167	153	134	279	244	214	421	353	310	641	547	423	854	706	550	1,164	977	705	1,451	1,188	860
100	175	163	NA	311	277	NA	489	421	NA	751	658	479	1,025	873	625	1,408	1,215	800	1,784	1,502	975

(continued)

Number of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

VENT CONNECTOR CAPACITY

VENT HEIGHT T (H) (feet)	CONNECTOR RISE (R) (feet)	TYPE B DOUBLE-WALL VENT AND DIAMETER—(D) inches																					
		12		14		16		18		20		22		24									
		APPLIANCE INPUT RATING LIMITS IN THOUSANDS OF BTU/H																					
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT				
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	
6	2	174	764	496	223	1,046	653	281	1,371	853	346	1,772	1,080	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	180	897	616	230	1,231	827	287	1,617	1,081	352	2,069	1,370	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	6	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
8	2	186	822	516	238	1,126	696	298	1,478	910	365	1,920	1,150	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	192	952	644	244	1,307	884	305	1,719	1,150	372	2,211	1,460	471	2,737	1,800	560	3,319	2,180	662	3,957	2,590	
	6	198	1,050	772	252	1,445	1,072	313	1,902	1,390	380	2,434	1,770	478	3,018	2,180	568	3,665	2,640	669	4,373	3,130	
10	2	196	870	536	249	1,195	730	311	1,570	955	379	2,049	1,205	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	201	997	664	256	1,371	924	318	1,804	1,205	387	2,332	1,535	486	2,887	1,890	581	3,502	2,280	686	4,175	2,710	
	6	207	1,095	792	263	1,509	1,118	325	1,989	1,455	395	2,556	1,865	494	3,169	2,290	589	3,849	2,760	694	4,593	3,270	
15	2	214	967	568	272	1,334	790	336	1,760	1,030	408	2,317	1,305	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	221	1,085	712	279	1,499	1,006	344	1,978	1,320	416	2,579	1,665	523	3,197	2,060	624	3,881	2,490	734	4,631	2,960	
	6	228	1,181	856	286	1,632	1,222	351	2,157	1,610	424	2,796	2,025	533	3,470	2,510	634	4,216	3,030	743	5,035	3,600	
20	2	223	1,051	596	291	1,443	840	357	1,911	1,095	430	2,533	1,385	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	230	1,162	748	298	1,597	1,064	365	2,116	1,395	438	2,778	1,765	554	3,447	2,180	661	4,190	2,630	772	5,005	3,130	
	6	237	1,253	900	307	1,726	1,288	373	2,287	1,695	450	2,984	2,145	567	3,708	2,650	671	4,511	3,190	785	5,392	3,790	
30	2	216	1,217	632	286	1,664	910	367	2,183	1,190	461	2,891	1,540	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	223	1,316	792	294	1,802	1,160	376	2,366	1,510	474	3,110	1,920	619	3,840	2,365	728	4,861	2,860	847	5,606	3,410	
	6	231	1,400	952	303	1,920	1,410	384	2,524	1,830	485	3,299	2,340	632	4,080	2,875	741	4,976	3,480	860	5,961	4,150	
50	2	206	1,479	689	273	2,023	1,007	350	2,659	1,315	435	3,548	1,665	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	213	1,561	860	281	2,139	1,291	359	2,814	1,685	447	3,730	2,135	580	4,601	2,633	709	5,569	3,185	851	6,633	3,790	
	6	221	1,631	1,031	290	2,242	1,575	369	2,951	2,055	461	3,893	2,605	594	4,808	3,208	724	5,826	3,885	867	6,943	4,620	
100	2	192	1,923	712	254	2,644	1,050	326	3,490	1,370	402	4,707	1,740	NA	NA	NA	NA	NA	NA	NA	NA	NA	
	4	200	1,984	888	263	2,731	1,346	336	3,606	1,760	414	4,842	2,220	523	5,982	2,750	639	7,254	3,330	769	8,650	3,950	
	6	208	2,035	1,064	272	2,811	1,642	346	3,714	2,150	426	4,968	2,700	539	6,143	3,350	654	7,453	4,070	786	8,892	4,810	

COMMON VENT CAPACITY

VENT HEIGHT (H) (feet)	TYPE B DOUBLE-WALL COMMON VENT DIAMETER—(D) inches																					
	12			14			16			18			20			22			24			
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																					
	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	
6	900	696	588	1,284	990	815	1,735	1,336	1,065	2,253	1,732	1,345	2,838	2,180	1,660	3,488	2,677	1,970	4,206	3,226	2,390	
8	994	773	652	1,423	1,103	912	1,927	1,491	1,190	2,507	1,936	1,510	3,162	2,439	1,860	3,890	2,998	2,200	4,695	3,616	2,680	
10	1,076	841	712	1,542	1,200	995	2,093	1,625	1,300	2,727	2,113	1,645	3,444	2,665	2,030	4,241	3,278	2,400	5,123	3,957	2,920	
15	1,247	986	825	1,794	1,410	1,158	2,440	1,910	1,510	3,184	2,484	1,910	4,026	3,133	2,360	4,971	3,862	2,790	6,016	4,670	3,400	
20	1,405	1,116	916	2,006	1,588	1,290	2,722	2,147	1,690	3,561	2,798	2,140	4,548	3,552	2,640	5,573	4,352	3,120	6,749	5,261	3,800	
30	1,658	1,327	1,025	2,373	1,892	1,525	3,220	2,558	1,990	4,197	3,326	2,520	5,303	4,193	3,110	6,539	5,157	3,680	7,940	6,247	4,480	

50	2,024	1,640	1,280	2,911	2,347	1,863	3,964	3,183	2,430	5,184	4,149	3,075	6,567	5,240	3,800	8,116	6,458	4,500	9,837	7,813	5,475
100	2,569	2,131	1,670	3,732	3,076	2,450	5,125	4,202	3,200	6,749	5,509	4,050	8,597	6,986	5,000	10,681	8,648	5,920	13,004	10,499	7,200

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

Number of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Single-wall metal connector

VENT CONNECTOR CAPACITY

VENT HEIGHT (H) (feet)	CONNECTOR RISE (R) (feet)	SINGLE-WALL METAL VENT CONNECTOR DIAMETER—(D) inches																									
		3		4			5			6			7			8			9			10					
		APPLIANCE INPUT RATING LIMITS IN THOUSANDS OF BTU/H																									
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT		
Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	
6	1	NA	NA	26	NA	NA	46	NA	NA	71	NA	NA	102	207	223	140	262	293	183	325	373	234	447	463	286		
	2	NA	NA	31	NA	NA	55	NA	NA	85	168	182	123	215	251	167	271	331	219	334	422	281	458	524	344		
	3	NA	NA	34	NA	NA	62	121	131	95	175	198	138	222	273	188	279	361	247	344	462	316	468	574	385		
8	1	NA	NA	27	NA	NA	48	NA	NA	75	NA	NA	106	226	240	145	285	316	191	352	403	244	481	502	299		
	2	NA	NA	32	NA	NA	57	125	126	89	184	193	127	234	266	173	293	353	228	360	450	292	492	560	355		
	3	NA	NA	35	NA	NA	64	130	138	100	191	208	144	241	287	197	302	381	256	370	489	328	501	609	400		
10	1	NA	NA	28	NA	NA	50	119	121	77	182	186	110	240	253	150	302	335	196	372	429	252	506	534	308		
	2	NA	NA	33	84	85	59	124	134	91	189	203	132	248	278	183	311	369	235	381	473	302	517	589	368		
	3	NA	NA	36	89	91	67	129	144	102	197	217	148	257	299	203	320	398	265	391	511	339	528	637	413		
15	1	NA	NA	29	79	87	52	116	138	81	177	214	116	238	291	158	312	380	208	397	482	266	556	596	324		
	2	NA	NA	34	83	94	62	121	150	97	185	230	138	246	314	189	321	411	248	407	522	317	568	646	387		
	3	NA	NA	39	87	100	70	127	160	109	193	243	157	255	333	215	331	438	281	418	557	360	579	690	437		
20	1	49	56	30	78	97	54	115	152	84	175	238	120	233	325	165	306	425	217	390	538	276	546	664	336		
	2	52	59	36	82	103	64	120	163	101	182	252	144	243	346	197	317	453	259	400	574	331	558	709	403		
	3	55	62	40	87	107	72	125	172	113	190	264	164	252	363	223	326	476	294	412	607	375	570	750	457		
30	1	47	60	31	77	110	57	112	175	89	169	278	129	226	380	175	296	497	230	378	630	294	528	779	358		
	2	51	62	37	81	115	67	117	185	106	177	290	152	236	397	208	307	521	274	389	662	349	541	819	425		
	3	54	64	42	85	119	76	122	193	120	185	300	172	244	412	235	316	542	309	400	690	394	555	855	482		
50	1	46	69	34	75	128	60	109	207	96	162	336	137	217	460	188	284	604	245	364	768	314	507	951	384		
	2	49	71	40	79	132	72	114	215	113	170	345	164	226	473	223	294	623	293	376	793	375	520	983	458		
	3	52	72	45	83	136	82	119	221	123	178	353	186	235	486	252	304	640	331	387	816	423	535	1,013	518		
100	1	45	79	34	71	150	61	104	249	98	153	424	140	205	585	192	269	774	249	345	993	321	476	1,236	393		
	2	48	80	41	75	153	73	110	255	115	160	428	167	212	593	228	279	788	299	358	1,011	383	490	1,259	469		
	3	51	81	46	79	157	85	114	260	129	168	433	190	222	603	256	289	801	339	368	1,027	431	506	1,280	527		

COMMON VENT CAPACITY

VENT HEIGHT (H) (feet)	TYPE B DOUBLE-WALL COMMON VENT DIAMETER— (D) inches																				
	4			5			6			7			8			9			10		
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																				
	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT
6	NA	78	64	NA	113	99	200	158	144	304	244	196	398	310	257	541	429	332	665	515	407
8	NA	87	71	NA	126	111	218	173	159	331	269	218	436	342	285	592	473	373	730	569	460
10	NA	94	76	163	137	120	237	189	174	357	292	236	467	369	309	638	512	398	787	617	487
15	121	108	88	189	159	140	275	221	200	416	343	274	544	434	357	738	599	456	905	718	553
20	131	118	98	208	177	156	305	247	223	463	383	302	606	487	395	824	673	512	1,013	808	626
30	145	132	113	236	202	180	350	286	257	533	446	349	703	570	459	958	790	593	1,183	952	723
50	159	145	128	268	233	208	406	337	296	622	529	410	833	686	535	1,139	954	689	1,418	1,157	838
100	166	153	NA	297	263	NA	469	398	NA	726	633	464	999	846	606	1,378	1,185	780	1,741	1,459	948

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

Number of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

VENT CONNECTOR CAPACITY

VENT HEIGHT (H) (feet)		CONNECTOR RISE (R) (feet)		TYPE B DOUBLE-WALL VENT CONNECTOR DIAMETER—(D) inches																							
				3		4		5		6		7		8		9		10									
				APPLIANCE INPUT RATING LIMITS IN THOUSANDS OF BTU/H																							
				FAN		NAT		FAN		NAT		FAN		NAT		FAN		NAT		FAN		NAT		FAN		NAT	
Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	
6	1	24	33	21	39	62	40	52	106	67	65	194	101	87	274	141	104	370	201	124	479	253	145	599	319		
	2	26	43	28	41	79	52	53	133	85	67	230	124	89	324	173	107	436	232	127	562	300	148	694	378		
	3	27	49	34	42	92	61	55	155	97	69	262	143	91	369	203	109	491	270	129	633	349	151	795	439		
8	1	24	39	22	39	72	41	55	117	69	71	213	105	94	304	148	113	414	210	134	539	267	156	682	335		
	2	26	47	29	40	87	53	57	140	86	73	246	127	97	350	179	116	473	240	137	615	311	160	776	394		
	3	27	52	34	42	97	62	59	159	98	75	269	145	99	383	206	119	517	276	139	672	358	163	848	452		
10	1	24	42	22	38	80	42	55	130	71	74	232	108	101	324	153	120	444	216	142	582	277	165	739	348		
	2	26	50	29	40	93	54	57	153	87	76	261	129	103	366	184	123	498	247	145	652	321	168	825	407		
	3	27	55	35	41	105	63	58	170	100	78	284	148	106	397	209	126	540	281	147	705	366	171	893	463		
15	1	24	48	23	38	93	44	54	154	74	72	277	114	100	384	164	125	511	229	153	658	297	184	824	375		
	2	25	55	31	39	105	55	56	174	89	74	299	134	103	419	192	128	558	260	156	718	339	187	900	432		
	3	26	59	35	41	115	64	57	189	102	76	319	153	105	448	215	131	597	292	159	760	382	190	960	486		
20	1	24	52	24	37	102	46	53	172	77	71	313	119	98	437	173	123	584	239	150	752	312	180	943	397		
	2	25	58	31	39	114	56	55	190	91	73	335	138	101	467	199	126	625	270	153	805	354	184	1,011	452		
	3	26	63	35	40	123	65	57	204	104	75	353	157	104	493	222	129	661	301	156	851	396	187	1,067	505		
30	1	24	54	25	37	111	48	52	192	82	69	357	127	96	504	187	119	680	255	145	883	337	175	1,115	432		
	2	25	60	32	38	122	58	54	208	95	72	376	145	99	531	209	122	715	287	149	928	378	179	1,171	484		
	3	26	64	36	40	131	66	56	221	107	74	392	163	101	554	233	125	746	317	152	968	418	182	1,220	535		
50	1	23	51	25	36	116	51	51	209	89	67	405	143	92	582	213	115	798	294	140	1,049	392	168	1,334	506		
	2	24	59	32	37	127	61	53	225	102	70	421	161	95	604	235	118	827	326	143	1,085	433	172	1,379	558		
	3	26	64	36	39	135	69	55	237	115	72	435	80	98	624	260	121	854	357	147	1,118	474	176	1,421	611		
100	1	23	46	24	35	108	50	49	208	92	65	428	155	88	640	237	109	907	334	134	1,222	454	161	1,589	596		
	2	24	53	31	37	120	60	51	224	105	67	444	174	92	660	260	113	933	368	138	1,253	497	165	1,626	651		
	3	25	59	35	38	130	68	53	237	118	69	458	193	94	679	285	116	956	399	141	1,282	540	169	1,661	705		

COMMON VENT CAPACITY

VENT HEIGHT (H) (feet)	MINIMUM INTERNAL AREA OF MASONRY CHIMNEY FLUE (square inches)																							
	12		19		28		38		50		63		78		113									
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																							
	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT
6	NA	74	25	NA	119	46	NA	178	71	NA	257	103	NA	351	143	NA	458	188	NA	582	246	1,041	853	NA
8	NA	80	28	NA	130	53	NA	193	82	NA	279	119	NA	384	163	NA	501	218	724	636	278	1,144	937	408
10	NA	84	31	NA	138	56	NA	207	90	NA	299	131	NA	409	177	606	538	236	776	686	302	1,226	1,010	454
15	NA	NA	36	NA	152	67	NA	233	106	NA	334	152	523	467	212	682	611	283	874	781	365	1,374	1,156	546
20	NA	NA	41	NA	NA	75	NA	250	122	NA	368	172	565	508	243	742	668	325	955	858	419	1,513	1,286	648
30	NA	NA	NA	NA	NA	NA	NA	270	137	NA	404	198	615	564	278	816	747	381	1,062	969	496	1,702	1,473	749
50	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	620	328	879	831	461	1,165	1,089	606	1,905	1,692	922
100	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	348	NA	NA	499	NA	NA	669	2,053	1,921	1,058

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

Numbr of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Single-wall metal connector

VENT CONNECTOR CAPACITY

VENT HEIGHT (H) (feet)		CONNECTOR RISE		SINGLE-WALL METAL VENT CONNECTOR DIAMETER (D) — inches																							
				3		4		5		6		7		8		9		10									
				APPLIANCE INPUT RATING LIMITS IN THOUSANDS OF BTU/H																							
		FAN		NAT		FAN		NAT		FAN		NAT		FAN		NAT		FAN		NAT		FAN		NAT			
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max		
6	1	NA	NA	21	NA	NA	39	NA	NA	66	179	191	100	231	271	140	292	366	200	362	474	252	499	594	316		
	2	NA	NA	28	NA	NA	52	NA	NA	84	186	227	123	239	321	172	301	432	231	373	557	299	509	696	376		
	3	NA	NA	34	NA	NA	61	134	153	97	193	258	142	247	365	202	309	491	269	381	634	348	519	793	437		

8	1	NA	NA	21	NA	NA	40	NA	NA	68	195	208	103	250	298	146	313	407	207	387	530	263	529	672	331
	2	NA	NA	28	NA	NA	52	137	139	85	202	240	125	258	343	177	323	465	238	397	607	309	540	766	391
	3	NA	NA	34	NA	NA	62	143	156	98	210	264	145	266	376	205	332	509	274	407	663	356	551	838	450
10	1	NA	NA	22	NA	NA	41	130	151	70	202	225	106	267	316	151	333	434	213	410	571	273	558	727	343
	2	NA	NA	29	NA	NA	53	136	150	86	210	255	128	276	358	181	343	489	244	420	640	317	569	813	403
	3	NA	NA	34	97	102	62	143	166	99	217	277	147	284	389	207	352	530	279	430	694	363	580	880	459
15	1	NA	NA	23	NA	NA	43	129	151	73	199	271	112	268	376	161	349	502	225	445	646	291	623	808	366
	2	NA	NA	30	92	103	54	135	170	88	207	295	132	277	411	189	359	548	256	456	706	334	634	884	424
	3	NA	NA	34	96	112	63	141	185	101	215	315	151	286	439	213	368	586	289	466	755	378	646	945	479
20	1	NA	NA	23	87	99	45	128	167	76	197	303	117	265	425	169	345	569	235	439	734	306	614	921	347
	2	NA	NA	30	91	111	55	134	185	90	205	325	136	274	455	195	355	610	266	450	787	348	627	986	443
	3	NA	NA	35	96	119	64	140	199	103	213	343	154	282	481	219	365	644	298	461	831	391	639	1,042	496
30	1	NA	NA	24	86	108	47	126	187	80	193	347	124	259	492	183	338	665	250	430	864	330	600	1,089	421
	2	NA	NA	31	91	119	57	132	203	93	201	366	142	269	518	205	348	699	282	442	908	372	613	1,145	473
	3	NA	NA	35	95	127	65	138	216	105	209	381	160	277	540	229	358	729	312	452	946	412	626	1,193	524
50	1	NA	NA	24	85	113	50	124	204	87	188	392	139	252	567	208	328	778	287	417	1,022	383	582	1,302	492
	2	NA	NA	31	89	123	60	130	218	100	196	408	158	262	588	230	339	806	320	429	1,058	425	596	1,346	545
	3	NA	NA	35	94	131	68	136	231	112	205	422	176	271	607	255	349	831	351	440	1,090	466	610	1,386	597
100	1	NA	NA	23	84	104	49	122	200	89	182	410	151	243	617	232	315	875	328	402	1,181	444	560	1,537	580
	2	NA	NA	30	88	115	59	127	215	102	190	425	169	253	636	254	326	899	361	415	1,210	488	575	1,570	634
	3	NA	NA	34	93	124	67	133	228	115	199	438	188	262	654	279	337	921	392	427	1,238	529	589	1,604	687

COMMON VENT CAPACITY

VENT HEIGHT	MINIMUM INTERNAL AREA OF MASONRY CHIMNEY FLUE (square inches)																								
	12			19			28			38			50			63			78			113			
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																								
	(H) (feet)	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT
6	NA	NA	NA	25	NA	118	45	NA	176	71	NA	255	102	NA	348	142	NA	455	187	NA	579	245	NA	846	NA
8	NA	NA	NA	28	NA	128	52	NA	190	81	NA	276	118	NA	380	162	NA	497	217	NA	633	277	1,136	928	405
10	NA	NA	NA	31	NA	136	56	NA	205	89	NA	295	129	NA	405	175	NA	532	234	171	680	300	1,216	1,000	450
15	NA	NA	NA	36	NA	NA	66	NA	230	105	NA	335	150	NA	400	210	677	602	280	866	772	360	1,359	1,139	540
20	NA	NA	NA	NA	NA	NA	74	NA	247	120	NA	362	170	NA	503	240	765	661	321	947	849	415	1,495	1,264	640
30	NA	NA	NA	NA	NA	NA	NA	NA	NA	135	NA	398	195	NA	558	275	808	739	377	1,052	957	490	1,682	1,447	740
50	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	612	325	NA	821	456	1,152	1,076	600	1,879	1,672	910
100	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	494	NA	NA	663	2,006	1,885	1,046

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

Number of Appliances	Two or more
Appliance Type	Draft hood-equipped
Appliance Vent Connection	Direct to pipe or vent

VENT CONNECTOR CAPACITY

TOTAL VENT HEIGHT (H) (feet)	CONNECTOR RISE (R) (feet)	VENT CONNECTOR DIAMETER—(D) inches					
		3	4	5	6	7	8
		MAXIMUM APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H					
6–8	1	21	40	68	102	146	205
	2	28	53	86	124	178	235
	3	34	61	98	147	204	275
15	1	23	44	77	117	179	240
	2	30	56	92	134	194	265
	3	35	64	102	155	216	298
30 and up	1	25	49	84	129	190	270
	2	31	58	97	145	211	295
	3	36	68	107	164	232	321

COMMON VENT CAPACITY

TOTAL VENT HEIGHT (H) (feet)	COMMON VENT DIAMETER—(D) inches						
	4	5	6	7	8	10	12
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H						
6	48	78	111	155	205	320	NA
8	55	89	128	175	234	365	505
10	59	95	136	190	250	395	560
15	71	115	168	228	305	480	690
20	80	129	186	260	340	550	790
30	NA	147	215	300	400	650	940
50	NA	NA	NA	360	490	810	1,190

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

⋮

Number of Appliances	Single
Appliance Type	Draft hood equipped
Appliance Vent Connection	Connected directly to pipe or vent

HEIGHT (H) (feet)	LATERAL (L) (feet)	VENT DIAMETER—(D) inches							
		3	4	5	6	7	8	10	12
		MAXIMUM APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H							
6	0	39	70	116	170	232	312	500	750
	2	31	55	94	141	194	260	415	620
	5	28	51	88	128	177	242	390	600
8	0	42	76	126	185	252	340	542	815
	2	32	61	102	154	210	284	451	680
	5	29	56	95	141	194	264	430	648
	10	24	49	86	131	180	250	406	625
10	0	45	84	138	202	279	372	606	912
	2	35	67	111	168	233	311	505	760
	5	32	61	104	153	215	289	480	724
	10	27	54	94	143	200	274	455	700
	15	NA	46	84	130	186	258	432	666
15	0	49	91	151	223	312	420	684	1,040
	2	39	72	122	186	260	350	570	865
	5	35	67	110	170	240	325	540	825
	10	30	58	103	158	223	308	514	795
	15	NA	50	93	144	207	291	488	760
	20	NA	NA	82	132	195	273	466	726
20	0	53	101	163	252	342	470	770	1,190
	2	42	80	136	210	286	392	641	990

	5	38	74	123	192	264	364	610	945
	10	32	65	115	178	246	345	571	910
	15	NA	55	104	163	228	326	550	870
	20	NA	NA	91	149	214	306	525	832
30	0	56	108	183	276	384	529	878	1,370
	2	44	84	148	230	320	441	730	1,140
	5	NA	78	137	210	296	410	694	1,080
	10	NA	68	125	196	274	388	656	1,050
	15	NA	NA	113	177	258	366	625	1,000
	20	NA	NA	99	163	240	344	596	960
	30	NA	NA	NA	NA	192	295	540	890
50	0	NA	120	210	310	443	590	980	1,550
	2	NA	95	171	260	370	492	820	1,290
	5	NA	NA	159	234	342	474	780	1,230
	10	NA	NA	146	221	318	456	730	1,190
	15	NA	NA	NA	200	292	407	705	1,130
	20	NA	NA	NA	185	276	384	670	1,080
	30	NA	NA	NA	NA	222	330	605	1,010

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

TABLE 504.2(6)
EXTERIOR MASONRY CHIMNEY

Number of Appliances	Single
Appliance Type	NAT
Appliance Vent Connection	Type B double-wall connector

MINIMUM ALLOWABLE INPUT RATING OF SPACE-HEATING APPLIANCE IN THOUSANDS OF BTU PER HOUR								
VENT HEIGHT (feet)	Internal area of chimney (square inches)							
	12	19	28	38	50	63	78	113
37°F or Greater Local 99% Winter Design Temperature: 37°F or Greater								
6	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0
15	NA	0	0	0	0	0	0	0
20	NA	NA	123	190	249	184	0	0
30	NA	NA	NA	NA	NA	393	334	0
50	NA	NA	NA	NA	NA	NA	NA	579
27 to 36°F Local 99% Winter Design Temperature: 27 to 36°F								
6	0	0	68	116	156	180	212	266
8	0	0	82	127	167	187	214	263
10	0	51	97	141	183	201	225	265
15	NA	NA	NA	NA	233	253	274	305

20	NA	NA	NA	NA	NA	307	330	362
30	NA	NA	NA	NA	NA	419	445	485
50	NA	NA	NA	NA	NA	NA	NA	763
17 to 26°F Local 99% Winter Design Temperature: 17 to 26°F								
6	NA	NA	NA	NA	NA	215	259	349
8	NA	NA	NA	NA	197	226	264	352
10	NA	NA	NA	NA	214	245	278	358
15	NA	NA	NA	NA	NA	296	331	398
20	NA	NA	NA	NA	NA	352	387	457
30	NA	NA	NA	NA	NA	NA	507	581
50	NA	NA	NA	NA	NA	NA	NA	NA
5 to 16°F Local 99% Winter Design Temperature: 5 to 16°F								
6	NA	NA	NA	NA	NA	NA	NA	416
8	NA	NA	NA	NA	NA	NA	312	423
10	NA	NA	NA	NA	NA	289	331	430
15	NA	NA	NA	NA	NA	NA	393	485
20	NA	NA	NA	NA	NA	NA	450	547
30	NA	NA	NA	NA	NA	NA	NA	682
50	NA	NA	NA	NA	NA	NA	NA	972
-10 to 4°F Local 99% Winter Design Temperature: -10 to 4°F								
6	NA	NA	NA	NA	NA	NA	NA	484
8	NA	NA	NA	NA	NA	NA	NA	494
10	NA	NA	NA	NA	NA	NA	NA	513
15	NA	NA	NA	NA	NA	NA	NA	586
20	NA	NA	NA	NA	NA	NA	NA	650
30	NA	NA	NA	NA	NA	NA	NA	805
50	NA	NA	NA	NA	NA	NA	NA	1,003
-11°F or Lower Local 99% Winter Design Temperature: -11°F or Lower								
Not recommended for any vent configurations								

For SI: °C = (°F - 32)/1.8, 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

504.2.3 Vent offsets. Single-appliance venting configurations with zero (0) lateral lengths in Tables 504.2(1), 504.2(2) and 504.2(5) shall not have elbows in the venting system. Single-appliance venting configurations with lateral lengths include two 90-degree (1.57 rad) elbows. For each additional elbow up to and including 45 degrees (0.79 rad), the maximum capacity listed in the venting tables shall be reduced by 5 percent. For each additional elbow greater than 45 degrees (0.79 rad) up to and including 90 degrees (1.57 rad), the maximum capacity listed in the venting tables shall be reduced by 10 percent. Where multiple offsets occur in a vent, the total lateral length of all offsets combined shall not exceed that specified in Tables 504.2(1) through 504.2(5).

504.2.4 Zero lateral. Zero (0) lateral (L) shall apply only to a straight vertical vent attached to a top outlet draft hood or flue collar.

504.2.5 Reserved.

504.2.6 Multiple input rate appliances. For appliances with more than one input rate, the minimum vent capacity (FAN Min) determined from the tables shall be less than the lowest appliance input rating, and the maximum vent capacity (FAN Max/NAT Max) determined from the tables shall be greater than the highest appliance rating input.

504.2.7 Liner system sizing and connections. Listed corrugated metallic chimney liner systems in masonry chimneys shall be sized by using Table 504.2(1) or 504.2(2) for Type B vents with the maximum capacity reduced by 20 percent ($0.80 \times$ maximum capacity) and the minimum capacity as shown in Table 504.2(1) or 504.2(2). Corrugated metallic liner systems installed with bends or offsets

shall have their maximum capacity further reduced in accordance with Section 504.2.3. The 20-percent reduction for corrugated metallic chimney liner systems includes an allowance for one long-radius 90-degree (1.57 rad) turn at the bottom of the liner.

Connections between chimney liners and listed double-wall connectors shall be made with listed adapters designed for such purpose.

504.2.8 Vent area and diameter. Where the vertical vent has a larger diameter than the vent connector, the vertical vent diameter shall be used to determine the minimum vent capacity, and the connector diameter shall be used to determine the maximum vent capacity. The flow area of the vertical vent shall not exceed seven times the flow area of the listed appliance categorized vent area, flue collar area or draft hood outlet area unless designed in accordance with approved engineering methods.

504.2.9 Chimney and vent locations. Tables 504.2(1), 504.2(2), 504.2(3), 504.2(4) and 504.2(5) shall be used only for chimneys and vents not exposed to the outdoors below the roof line. Table 504.2(3) in combination with Table 504.2(6) shall be used for clay-tile-lined exterior masonry chimneys, provided that all of the following are met:

1. Vent connector is a Type B double-wall.
2. Vent connector length is limited to 1½ feet for each inch (18 mm for each mm) of vent connector diameter.
3. The appliance is draft hood equipped.
4. The input rating is less than the maximum capacity given by Table 504.2(3).
5. For a water heater, the outdoor design temperature is not less than 5°F (-15°C).
6. For a space-heating appliance, the input rating is greater than the minimum capacity given by Table 504.2(6).

Where these conditions cannot be met, an alternative venting design shall be used, such as a listed chimney lining system.

Exception: The installation of vents serving listed appliances shall be permitted to be in accordance with the appliance manufacturer's instructions and the terms of the listing.

504.2.10 Corrugated vent connector size. Corrugated vent connectors shall be not smaller than the listed appliance categorized vent diameter, flue collar diameter or draft hood outlet diameter.

504.2.11 Vent connector size limitation. Vent connectors shall not be increased in size more than two sizes greater than the listed appliance categorized vent diameter, flue collar diameter, or draft hood outlet diameter.

504.2.12 Component commingling. In a single run of vent or vent connector, different diameters and types of vent and connector components shall be permitted to be used, provided that all such sizes and types are permitted by the tables.

504.2.13 Draft hood conversion accessories. Draft hood conversion accessories for use with masonry chimneys venting listed Category I fan-assisted appliances shall be listed and installed in accordance with the manufacturer's instructions for such listed accessories.

504.2.14 Table interpolation. Interpolation shall be permitted in calculating capacities for vent dimensions that fall between the table entries.

504.2.15 Extrapolation prohibited. Extrapolation beyond the table entries shall not be permitted.

504.2.16 Engineering calculations. For vent heights less than 6 feet (1828.8 mm) and greater than shown in the tables, engineering methods shall be used to calculate vent capacities. Signed and sealed calculations shall be submitted with filing application.

504.2.17 Height entries. Where the actual height of a vent falls between entries in the height column of the applicable table in Tables 504.2(1) through 504.2(6), either interpolation shall be used or the lower appliance input rating shown in the table entries shall be used for FAN MAX and NAT MAX column values and the higher appliance input rating shall be used for the FAN MIN column values.

504.3 Application of multiple appliance vent Tables 504.3(1) through 504.3(7b). The application of Tables 504.3(1) through 504.3(7) shall be subject to the requirements of Sections 504.3.1 through 504.3.28.

TABLE 504.3(1)
TYPE B DOUBLE-WALL VENT

Number of Appliances	Single
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

**VENT CONNECTOR
CAPACITY**

VENT HEIG HT (H) (feet)	CONNEC TOR RISE (R) (feet)	TYPE B DOUBLE-WALL VENT AND CONNECTOR DIAMETER – (D) inches																							
		3		4		5		6		7		8		9		10									
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																							
		FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NAT	FAN		NA T	FAN		NA T	FAN		N A T
		Mi n	M ax	Ma x	Mi n	M ax	Ma x	Mi n	M ax	Ma x	Mi n	M ax	Ma x	Mi n	M ax	Ma x	Mi n	M ax	Ma x	Mi n	M ax	Ma x	Mi n	M ax	Ma x
6	1	22	37	26	35	66	46	46	106	72	58	164	104	77	225	142	92	296	185	109	376	237	128	466	289
	2	23	41	31	37	75	55	48	121	86	60	183	124	79	253	168	95	333	220	112	424	282	131	526	345
	3	24	44	35	38	81	62	49	132	96	62	199	139	82	275	189	97	363	248	114	463	317	134	575	386
8	1	22	40	27	35	72	48	49	114	76	64	176	109	84	243	148	100	320	194	118	408	248	138	507	303
	2	23	44	32	36	80	57	51	128	90	66	195	129	86	269	175	103	356	230	122	454	294	141	564	355

																				1					8
	3	24	47	36	37	87	64	53	13 9	10 1	67	21 0	14 5	88	29 0	198	10 5	384	258	1 2 3	492	330	143	612	4 0 2
10	1	22	43	28	34	78	50	49	12 3	78	65	18 9	11 3	89	25 7	154	10 6	341	200	1 2 5	436	257	146	542	3 1 4
	2	23	47	33	36	86	59	51	13 6	93	67	20 6	13 4	91	28 2	182	10 9	374	238	1 2 8	479	305	149	596	3 7 2
	3	24	50	37	37	92	67	52	14 6	10 4	69	22 0	15 0	94	30 3	205	11 1	402	268	1 3 1	515	342	152	642	4 1 7
15	1	21	50	30	33	89	53	47	14 2	83	64	22 0	12 0	88	29 8	163	11 0	389	214	1 3 4	493	273	162	609	3 3 3
	2	22	53	35	35	96	63	49	15 3	99	66	23 5	14 2	91	32 0	193	11 2	419	253	1 3 7	532	323	165	658	3 9 4
	3	24	55	40	36	10 2	71	51	16 3	11 1	68	24 8	16 0	93	33 9	218	11 5	445	286	1 4 0	565	365	167	700	4 4 4
20	1	21	54	31	33	99	56	46	15 7	87	62	24 6	12 5	86	33 4	171	10 7	436	224	1 3 1	552	285	158	681	3 4 7
	2	22	57	37	34	10 5	66	48	16 7	10 4	64	25 9	14 8	89	35 4	202	11 0	463	265	1 3 4	587	339	161	725	4 1 4
	3	23	60	42	35	11 0	74	50	17 6	11 6	66	27 1	16 8	91	37 1	228	11 3	486	300	1 3 7	618	383	164	764	4 6 6
30	1	20	62	33	31	11 3	59	45	18 1	93	60	28 8	13 4	93	39 1	182	10 3	512	238	1 2 5	649	305	151	802	3 7 2

	2	21	64	39	33	118	70	47	190	110	62	299	158	95	408	215	105	535	282	129	679	360	155	840	439
	3	22	66	44	34	123	79	48	198	124	64	309	178	99	423	242	108	555	317	132	706	405	158	874	494
50	1	19	71	36	30	133	64	43	216	101	57	349	145	89	477	197	97	627	257	120	797	330	144	984	403
	2	21	73	43	32	137	76	45	223	119	59	358	172	91	490	234	100	645	306	123	820	392	148	1,014	478
	3	22	75	48	33	141	86	46	229	134	61	366	194	93	502	263	103	661	3434	126	842	441	151	1,043	538
100	1	18	82	37	28	158	66	40	262	104	53	442	150	73	611	204	91	810	266	112	1,038	341	135	1,285	417
	2	19	83	44	30	161	79	42	267	123	55	447	178	75	619	242	94	822	316	115	1,054	405	139	1,306	494
	3	20	84	50	31	163	89	44	272	138	57	452	109	78	627	272	97	834	955	118	1,069	455	142	1,327	555

COMMON VENT CAPACITY

[illegible]

6	92	81	65	140	116	103	204	161	147	309	248	200	404	314	260	547	434	335	672	520	410
8	101	90	73	155	129	114	224	178	163	339	275	223	444	348	290	602	480	378	740	577	465
10	110	97	79	169	141	124	243	194	178	367	299	242	477	377	315	649	522	405	800	627	495
15	125	112	91	195	164	144	283	228	206	427	352	280	556	444	365	753	612	465	924	733	565
20	136	123	102	215	183	160	314	255	229	475	394	310	621	499	405	842	688	423	1,035	826	640
30	152	138	118	244	210	185	361	297	266	547	459	360	720	585	470	979	808	605	1,209	975	740
50	167	153	134	279	244	214	421	353	310	641	547	423	854	706	550	1,164	977	705	1,451	1,188	860
100	175	163	NA	311	277	NA	489	421	NA	751	658	479	1,025	873	625	1,408	1,215	800	1,784	1,502	975

(continued)

**TABLE 504.3(1)-continued
TYPE B DOUBLE-WALL VENT**

Number of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

VENT CONNECTOR CAPACITY

VENT HEIG HT (H) (feet)	CONNEC TOR RISE (R) (feet)																					
		12		14		16		18		20		22		24								
		APPLIANCE INPUT RATING LIMITS IN THOUSANDS OF BTU/H																				
		FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T	FAN		NA T
		Min	Max	Ma x	Mi n	Ma x	Ma x	Mi n	Ma x	Ma x	Mi n	Ma x	Ma x	Mi n	Ma x	Ma x	Mi n	Max	Ma x	Mi n	Ma x	Ma x
6	2	174	764	496	22 3	1,0 46	563	28 1	1,3 71	853	34 5	1,7 32	1,0 80	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	180	897	616	23 0	1,2 31	827	28 7	1,6 17	1,0 81	35 2	2,0 69	1,3 70	N A	NA	NA	N A	NA	NA	N A	NA	NA
	6	NA	N A	NA	N A	NA	NA	N A	NA	NA	N A	NA	NA	N A	NA	NA	N A	NA	NA	N A	NA	NA
8	2	186	822	516	23 8	1,1 26	696	29 8	1,4 78	910	36 5	1,9 20	1,1 50	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	192	952	644	21 4	1,3 07	884	30 5	1,7 19	1,1 50	37 2	2,2 11	1,4 60	47 1	2,7 37	1,8 00	56 0	3,31 9	2,1 80	66 2	3,9 57	2,5 90

CHIMNEYS AND VENTS

	6	198	1,05 0	772	25 2	1,4 45	1,0 72	31 3	1,9 02	1,3 90	38 0	2,4 34	1,8 70	47 8	3,0 18	2,1 80	56 8	3,66 5	2,6 40	66 8	4,3 73	3,1 30
10	2	196	870	536	24 9	1,1 95	730	31 1	1,5 70	955	37 9	2,0 49	1,2 05	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	201	997	664	25 6	1,3 71	924	31 8	1,8 04	1,2 05	38 7	2,3 32	1,5 35	48 6	2,8 87	1,8 90	58 1	3,50 2	2,2 80	68 6	4,1 75	2,7 10
	6	207	1,09 5	792	26 3	1,5 09	1,1 18	32 5	1,9 89	1,4 55	39 5	2,5 56	1,8 65	49 4	2,3 09	2,2 90	58 9	3,84 9	2,7 60	69 4	4,5 93	3,2 70
15	2	214	967	568	23 2	1,3 51	790	33 6	1,7 60	1,2 05	40 8	2,3 17	1,3 05	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	221	1,08 5	715	27 9	1,4 99	1,0 06	34 4	1,9 78	1,3 20	41 6	2,5 79	1,6 65	52 3	3,1 97	2,0 50	62 4	3,88 1	2,4 90	73 4	4,6 31	2,9 60
	6	228	1,18 1	856	28 6	1,6 32	1,2 22	35 1	2,1 57	1,6 10	42 4	2,7 96	2,0 25	53 3	3,4 70	2,5 10	63 4	4,21 6	3,0 90	74 3	5,0 35	3,6 00
20	2	223	1,05 1	996	29 1	1,4 63	840	35 7	1,9 11	1,0 95	43 0	2,5 33	1,3 85	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	230	1,16 2	748	29 8	1,5 97	1,0 54	36 5	2,1 16	1,3 95	43 8	2,7 78	1,7 65	55 4	3,4 47	2,1 80	65 1	4,19 0	2,6 30	77 2	5,0 05	3,1 90
	6	237	1,25 3	900	30 7	1,7 26	1,2 88	37 3	2,2 87	1,6 95	45 0	2,9 84	2,1 45	56 7	3,7 08	2,6 90	67 1	4,51 1	3,1 90	78 5	5,3 92	3,7 90
30	2	216	1,21 7	632	28 6	1,6 64	910	36 7	2,1 83	1,1 90	46 1	2,8 91	1,3 40	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	223	1,31 6	792	29 4	1,8 02	1,1 60	37 6	2,3 66	1,5 10	47 4	3,1 10	1,9 20	61 9	3,8 40	2,3 65	72 8	4,86 1	2,8 60	84 7	5,6 06	3,4 10
	6	231	1,40 0	952	30 3	1,9 20	1,4 10	38 4	2,5 24	1,8 30	48 5	3,2 99	2,3 40	63 2	4,0 80	2,8 75	74 0	4,97 6	3,4 80	86 0	5,9 51	4,1 50

CHIMNEYS AND VENTS

50	2	206	1,47 9	689	27 3	2,0 23	1,0 07	35 0	2,6 59	1,3 15	43 5	3,5 48	1,6 65	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	213	1,56 1	860	28 1	2,1 99	1,2 91	35 9	2,8 14	1,6 85	44 7	3,7 30	2,1 35	58 0	4,6 01	2,6 33	70 9	5,56 0	3,1 85	85 1	6,6 33	3,7 90
	6	221	1,63 1	1,0 31	29 0	2,2 42	1,5 75	36 9	2,9 51	1,0 55	46 1	3,8 93	2,6 05	59 4	4,8 08	3,2 08	72 4	5,82 0	3,8 85	86 7	6,9 43	4,6 20
100	2	192	1,92 3	712	25 4	2,6 44	1,0 50	32 6	3,4 90	1,3 70	40 2	4,7 07	1,7 40	N A	NA	NA	N A	NA	NA	N A	NA	NA
	4	200	1,98 4	888	26 3	2,7 31	1,3 46	33 6	3,6 06	1,7 60	41 4	4,8 42	2,2 20	52 3	5,9 82	2,7 50	63 9	7,25 4	3,3 30	76 9	8,6 30	3,9 50
	6	208	2,03 5	1,0 64	27 2	2,8 11	1,6 42	34 6	3,7 14	2,1 50	42 6	4,9 68	2,7 00	59 9	6,1 43	3,3 50	65 4	7,45 3	4,0 70	78 6	8,8 92	4,8 10

CHIMNEYS AND VENTS

COMMON VENT CAPACITY

VENT HEIGHT (H) (feet)	TYPE B DOUBLE-WALL VENT AND CONNECTOR DIAMETER – (D) inches																				
	12			14			16			18			20			22			24		
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																				
	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT
6	900	696	588	1,284	990	815	1,735	1,336	1,065	2,253	1,732	1,345	2,838	2,180	1,660	3,488	2,677	1,970	4,205	3,226	2,390
8	994	773	652	1,423	1,103	912	1,927	1,491	1,190	2,507	1,936	1,510	3,162	2,439	1,860	3,890	2,998	2,200	4,695	3,616	2,680
10	1,076	941	712	1,542	1,200	995	2,093	1,625	1,500	2,727	2,113	1,645	3,444	2,665	2,030	4,241	3,278	2,400	5,123	3,957	2,920
15	1,247	986	825	1,794	1,410	1,158	2,440	1,910	1,510	3,184	2,484	1,910	4,026	3,133	2,160	4,971	3,862	2,790	6,016	4,670	9,400
20	1,405	1,116	916	2,006	1,588	1,290	2,722	2,147	1,690	3,561	2,798	2,140	4,548	3,552	2,640	5,573	4,332	3,120	6,749	5,231	3,800
30	1,658	1,325	1,025	2,373	1,892	1,525	3,220	2,558	1,990	4,197	3,326	2,520	5,303	4,193	3,110	6,539	5,157	3,680	7,940	6,247	4,430
50	2,024	1,640	1,280	2,911	2,347	1,863	3,964	3,183	2,430	5,184	4,149	3,075	6,567	5,240	3,800	8,116	6,458	4,500	9,837	7,813	5,475
100	2,569	2,131	1,670	3,732	3,076	2,450	5,125	4,202	3,200	6,749	5,509	4,050	5,897	6,986	5,000	10,681	8,648	5,920	13,704	10,439	7,200

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

CHIMNEYS AND VENTS

TABLE 504.3(2)
TYPE B DOUBLE-WALL VENT

Number of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Single-wall metal connector

VENT CONNECTOR
CAPACITY

VENT HEIGHT (H) (feet)	CONNECTOR RISE (R) (feet)	SINGLE-WALL METAL VENT CONNECTOR DIAMETER – (D) inches																							
		3			4			5			6			7			8			9			10		
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																							
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT
		Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max
6	1	NA	NA	26	NA	NA	46	NA	NA	71	NA	NA	102	207	223	140	262	293	183	325	373	234	447	463	286
	2	NA	NA	31	NA	NA	55	NA	NA	85	168	182	123	215	251	167	271	331	219	334	422	287	458	524	344
	3	NA	NA	34	NA	NA	62	NA	131	93	175	198	138	222	273	188	279	361	247	344	462	316	468	574	385
8	1	NA	NA	27	NA	NA	48	NA	NA	75	NA	NA	106	226	240	145	285	316	191	352	403	244	481	502	299
	2	NA	NA	32	NA	NA	57	125	126	89	181	193	127	234	266	173	293	353	228	360	450	292	492	560	355
	3	NA	NA	35	NA	NA	64	130	138	100	181	208	144	241	287	197	302	381	256	370	489	328	501	600	400
10	1	NA	NA	28	NA	NA	50	119	121	77	182	186	110	260	253	150	302	335	196	372	429	252	506	534	308

CHIMNEYS AND VENTS

	2	NA	NA	33	84	85	59	124	134	91	189	203	132	248	278	183	311	369	235	381	473	302	517	589	368
	3	NA	NA	36	89	91	67	129	144	102	197	217	148	258	299	203	320	398	255	391	511	339	528	637	413
15	1	NA	NA	29	79	87	52	116	138	81	177	214	116	238	291	158	312	380	208	397	452	266	556	596	324
	2	NA	NA	34	85	94	62	121	150	97	185	230	138	246	314	189	321	411	248	407	522	317	568	646	387
	3	NA	NA	39	87	100	70	127	160	109	193	243	157	255	333	215	831	438	281	418	557	360	579	690	437
20	1	49	56	30	78	97	54	115	152	84	175	238	120	233	323	165	306	425	217	390	539	276	546	664	336
	2	52	59	36	82	103	64	120	163	101	182	252	144	243	306	191	317	453	259	400	574	331	558	709	403
	3	55	62	40	87	107	72	125	172	113	190	264	164	252	363	223	326	476	294	412	607	375	570	750	457
30	1	47	60	31	77	110	57	112	175	89	169	278	129	226	380	175	296	497	230	378	630	294	528	779	358
	2	51	62	37	81	115	67	117	185	106	177	290	152	236	397	208	307	521	274	389	662	349	541	817	425
	3	54	64	42	85	119	76	122	193	120	185	300	172	244	412	235	316	542	309	400	690	394	555	855	482
50	1	46	69	34	75	128	60	109	207	96	162	336	137	217	460	188	284	604	245	364	768	314	507	951	384
	2	49	71	40	79	132	72	114	215	113	170	345	164	226	473	223	294	523	293	376	793	375	520	983	458
	3	52	72	45	83	136	82	119	221	123	178	353	186	235	486	252	304	640	331	387	816	423	535	1,013	518
100	1	45	79	34	71	150	61	104	249	98	153	424	100	205	585	192	269	714	240	345	993	321	476	1,236	393
	2	48	80	41	75	153	73	110	255	115	160	428	167	212	593	228	279	758	299	358	1,011	383	490	1,259	469
	3	51	81	46	79	157	85	114	260	129	168	433	190	222	603	256	289	807	339	368	1,027	431	506	1,280	527

CHIMNEYS AND VENTS

COMMON VENT CAPACITY

VENT HEIGHT (H) (feet)	TYPE B DOUBLE-WALL VENT AND CONNECTOR DIAMETER – (D) inches																					
	4			5			6			7			8			9			10			
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																					
	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	
6	NA	78	64	NA	115	99	200	158	144	304	244	196	398	310	257	541	429	332	665	515	407	
8	NA	87	71	NA	126	111	218	173	159	331	269	218	436	342	285	592	473	373	730	569	460	
10	NA	94	76	163	137	120	237	189	174	357	292	236	467	369	309	638	512	398	787	617	487	
15	121	108	88	189	159	140	275	221	200	416	343	274	544	434	357	738	599	456	905	718	553	
20	131	138	98	208	177	156	305	247	223	463	383	302	606	487	395	824	673	512	1,013	808	626	
30	145	132	113	236	202	180	350	286	257	535	446	349	703	570	459	958	790	593	1,183	952	723	
50	159	145	128	268	233	208	406	337	296	622	527	410	833	686	535	1,139	954	689	1,418	1,157	838	
100	166	153	NA	297	263	NA	469	398	NA	726	603	464	999	846	606	1,378	1,185	780	1,741	1,459	948	

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

CHIMNEYS AND VENTS

**TABLE 504.3(3)
MASONRY CHIMNEY**

Number of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Type B double-wall connector

**VENT CONNECTOR
CAPACITY**

VENT HEIGHT (H) (feet)	CONNECTOR RISE (R) (feet)	TYPE B DOUBLE-WALL VENT CONNECTOR DIAMETER – (D) inches																									
		3		4		5		6		7		8		9		10											
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																									
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT		
		Min	Max	Max	Min	Max	Ma40x52	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max		
6	1	24	33	21	39	62	40	52	106	67	65	194	101	87	274	141	104	370	201	124	479	253	145	599	319		
	2	26	43	28	41	79	52	53	135	85	67	230	124	89	324	173	107	436	232	127	562	300	148	694	378		
	3	27	49	34	42	92	61	55	155	97	69	262	143	91	369	200	109	491	270	129	533	349	151	795	439		
8	1	24	39	22	39	72	41	55	117	69	71	213	105	94	304	148	113	414	210	134	539	267	156	682	335		
	2	26	47	29	40	87	53	57	140	16	73	246	127	97	350	179	116	473	240	137	615	311	160	776	394		
	3	27	52	34	42	97	62	59	159	98	75	269	145	99	383	206	119	517	276	139	572	358	163	848	452		
10	1	24	42	22	38	80	42	55	130	71	74	232	108	101	324	153	120	444	216	142	582	277	165	739	348		
	2	26	50	29	40	93	54	57	153	87	76	264	129	103	366	184	123	498	247	145	652	321	168	825	407		

CHIMNEYS AND VENTS

	3	27	55	35	41	105	63	58	170	100	78	284	148	106	397	209	126	540	281	147	705	366	171	893	463
15	1	24	48	23	38	93	44	54	154	74	72	277	114	100	384	161	125	511	229	153	658	297	184	824	375
	2	25	55	31	39	105	55	56	174	89	74	299	134	103	419	182	128	558	260	156	718	339	187	900	452
	3	26	59	35	41	115	64	57	189	102	76	319	153	105	448	215	131	597	292	189	760	382	180	960	486
20	1	24	52	24	37	102	46	53	172	77	71	313	119	98	437	173	123	584	239	150	752	312	180	943	397
	2	25	58	31	39	114	56	55	190	91	73	335	138	101	467	199	126	625	270	153	805	354	184	1,011	452
	3	26	63	35	40	123	65	57	204	104	75	353	157	104	493	222	129	661	301	156	851	396	187	1,067	505
30	1	24	54	25	37	111	48	52	182	82	69	357	127	96	504	187	119	680	255	145	883	337	175	1,115	432
	2	25	60	32	38	132	58	54	208	95	72	376	145	99	531	209	122	715	287	149	928	378	179	1,171	484
	3	26	64	36	40	131	66	56	221	107	74	392	163	101	554	233	125	746	317	152	968	418	182	1,220	535
50	1	23	51	25	36	116	51	51	209	89	67	408	143	92	582	213	115	798	294	140	1,049	392	168	1,334	506
	2	24	59	32	37	127	61	53	225	102	70	421	161	95	604	235	118	827	326	143	1,085	433	172	1,379	558
	3	26	64	36	39	135	69	55	237	115	72	435	80	98	624	260	121	854	357	147	1,118	474	176	1,421	611
100	1	23	46	24	35	108	50	49	208	92	65	428	155	88	640	237	109	907	334	134	1,222	454	161	1,589	596
	2	24	53	31	37	120	60	51	224	105	67	444	174	92	660	260	113	933	368	138	1,253	497	165	1,626	651
	3	25	59	35	38	130	68	53	237	118	69	458	193	94	679	285	116	936	399	141	1,282	540	169	1,661	705

COMMON VENT CAPACITY

VENT	MINIMUM INTERNAL AREA OF MASONRY CHIMNEY FLUE (square inches)
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CHIMNEYS AND VENTS

HEIGHT (H) (feet)	12			19			28			38			50			63			78			113		
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																							
	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT
6	NA	74	25	NA	119	46	NA	178	71	NA	257	103	NA	351	143	NA	458	188	NA	582	246	1,041	853	NA
8	NA	80	28	NA	130	53	NA	193	82	NA	279	119	NA	384	163	NA	501	218	724	636	278	1,144	937	408
10	NA	84	31	NA	138	56	NA	207	90	NA	299	131	NA	409	177	605	538	236	776	686	302	1,226	1,010	454
15	NA	NA	36	NA	152	67	NA	233	106	NA	334	152	523	467	212	682	611	283	874	781	365	1,374	1,156	546
20	NA	NA	41	NA	NA	75	NA	250	122	NA	368	172	565	508	243	743	668	325	955	858	419	1,513	1,286	648
30	NA	NA	NA	NA	NA	NA	NA	270	137	NA	404	198	615	564	278	816	747	381	1,062	969	496	1,702	1,473	749
50	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	620	328	879	831	461	1,165	1,039	606	1,900	1,692	922
100	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	348	NA	NA	499	NA	NA	669	1,053	1,921	1,058

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 foot = 304.8mm, 1 British thermal unit per hour = 0.2931 W.

CHIMNEYS AND VENTS

TABLE 504.3(4)
MASONRY CHIMNEY

Number of Appliances	Two or more
Appliance Type	Category I
Appliance Vent Connection	Single-wall metal connector

VENT CONNECTOR
CAPACITY

VENT HEIGHT (H) (feet)	CONNECTOR RISE (R) (feet)	SINGLE-WALL METAL VENT CONNECTOR DIAMETER – (D) inches																									
		3		4			5			6			7			8			9			10					
		APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																									
		FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT	FAN		NAT		
		Min	Max	Max	Min	Max	Ma40x52	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max	Min	Max	Max		
6	1	NA	NA	21	NA	NA	39	NA	NA	66	179	191	100	231	271	140	292	365	200	362	474	252	499	594	316		
	2	NA	NA	28	NA	NA	52	NA	NA	84	136	227	123	239	321	172	301	432	231	373	557	299	509	696	376		
	3	NA	NA	34	NA	NA	61	134	153	97	193	258	142	247	365	202	309	491	263	381	634	348	519	703	437		
8	1	NA	NA	21	NA	NA	40	NA	NA	68	195	108	103	250	298	146	313	407	207	387	530	263	529	672	331		
	2	NA	NA	28	NA	NA	52	137	139	85	202	240	125	258	343	177	323	465	238	397	607	309	540	766	391		
	3	NA	NA	34	NA	NA	62	143	156	98	210	264	145	266	376	205	332	509	274	407	663	356	551	838	450		
10	1	NA	NA	22	NA	NA	41	130	151	70	202	225	106	267	316	151	333	434	213	410	571	273	558	727	343		
	2	NA	NA	29	NA	NA	53	136	150	86	210	255	128	276	358	181	346	489	244	420	640	317	569	813	403		

CHIMNEYS AND VENTS

	3	NA	NA	34	97	102	62	143	166	99	217	277	147	284	389	207	352	530	279	430	694	363	580	880	459
15	1	NA	NA	23	NA	NA	43	129	151	73	199	271	112	268	376	161	349	502	225	445	646	291	623	808	366
	2	NA	NA	60	92	103	54	135	170	88	207	295	132	277	411	189	359	548	256	456	706	334	634	884	424
	3	NA	NA	34	96	112	63	141	185	101	215	315	151	286	439	213	358	585	289	466	735	378	646	945	479
20	1	NA	NA	23	87	99	45	128	167	76	197	303	117	265	425	169	345	569	235	439	734	306	614	920	347
	2	NA	NA	30	91	111	55	134	185	90	205	325	136	274	455	195	355	610	266	450	787	348	627	986	443
	3	NA	NA	35	96	119	64	140	199	103	213	343	154	282	481	219	363	644	298	461	831	391	639	1,042	496
30	1	NA	NA	24	86	108	47	126	187	80	193	347	124	259	402	183	338	665	250	430	864	330	600	1,089	421
	2	NA	NA	31	91	119	57	132	203	93	201	366	142	269	518	205	348	699	282	442	908	372	613	1,145	473
	3	NA	NA	35	95	127	65	138	216	105	209	381	160	277	540	229	358	729	312	452	946	412	626	1,193	524
50	1	NA	NA	24	85	113	50	124	204	87	188	392	139	252	567	208	328	778	287	417	1,022	383	582	1,302	492
	2	NA	NA	31	89	123	60	130	218	100	196	408	158	262	588	230	339	805	320	429	1,058	425	595	1,346	545
	3	NA	NA	35	94	131	68	136	231	112	205	422	176	271	607	255	349	731	351	440	1,090	456	610	1,386	587
100	1	NA	NA	23	84	104	49	122	200	89	182	410	151	243	617	232	315	875	328	402	1,181	444	560	1,537	580
	2	NA	NA	30	88	115	59	127	215	102	190	425	169	253	636	254	326	899	361	415	1,200	438	575	1,570	634
	3	NA	NA	34	93	124	67	133	228	115	199	438	188	262	654	279	337	921	392	427	1,238	529	589	1,604	687

COMMON VENT CAPACITY

VENT	MINIMUM INTERNAL AREA OF MASONRY CHIMNEY FLUE (square inches)
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HEIGHT (H) (feet)	12			19			28			38			50			63			78			113			
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H																								
	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	FAN +FAN	FAN +NAT	NAT +NAT	
6	NA	NA	25	NA	118	45	NA	176	71	NA	255	102	NA	348	142	NA	455	187	NA	579	245	NA	846	NA	
8	NA	NA	28	NA	128	52	NA	150	81	NA	275	118	NA	380	162	NA	497	217	NA	633	277	1,136	928	405	
10	NA	NA	31	NA	136	56	NA	205	89	NA	295	129	NA	405	175	NA	532	234	171	680	300	1,216	1,000	450	
15	NA	NA	36	NA	NA	65	NA	230	105	NA	335	150	NA	400	210	677	602	280	866	772	360	1,359	1,139	540	
20	NA	NA	NA	NA	NA	74	NA	247	120	NA	362	170	NA	503	240	765	661	321	947	849	415	1,495	1,264	640	
30	NA	NA	NA	NA	NA	NA	NA	NA	135	NA	398	195	NA	558	275	808	739	377	1,052	957	490	1,682	1,447	740	
50	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	612	325	NA	NA	456	1,152	1,076	600	1,879	1,672	910	
100	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	494	NA	NA	663	2,006	1,835	1,046	

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 foot = 304.8mm, 1 British thermal unit per hour = 0.2931 W.

**TABLE 504.3(5)
SINGLE-WALL METAL PIPE
OR TYPE ASBESTOS
CEMENT VENT**

Number of Appliances	Two or more
Appliance Type	Draft hood-equipped
Appliance Vent Connection	Direct to pipe or vent

VENT CONNECTION CAPACITY

TOTAL VENT HEIGHT (H) (feet)	CONNECTOR RISE (R) (feet)	VENT CONNECTOR DIAMETER – (D) inches					
		3	4	5	6	7	8
		MAXIMUM APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H					
6-8	1	21	40	68	102	146	205
	2	28	53	86	124	178	235
	3	34	61	98	147	204	275
15	1	23	44	77	117	179	240
	2	30	56	92	134	194	265
	3	35	61	102	155	216	298
30 and up	1	25	49	84	129	190	270
	2	31	58	97	145	211	295
	3	36	68	107	164	232	321

COMMON VENT CAPACITY

TOTAL VENT HEIGHT (H) (feet)	COMMON VENT DIAMETER – (D) inches						
	4	5	6	7	8	10	12
	COMBINED APPLIANCE INPUT RATING IN THOUSANDS OF BTU/H						
6	48	78	111	155	205	320	NA

8	55	89	128	175	234	365	505
10	59	95	136	190	250	395	560
15	71	115	168	228	305	480	690
20	80	129	186	260	340	550	790
30	NA	147	215	300	400	650	940
50	NA	NA	NA	360	490	810	1,190

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

TABLE 504.3(6a)
EXTERIOR MASONRY CHIMNEY

Number of Appliances	Two or more
Appliance Type	NAT + NAT
Appliance Vent Connection	Type B double-wall connector

**Combined Appliance Maximum
Input Rating in Thousands of Btu per Hour**

VENT HEIGHT (feet)	INTERNAL AREA OF CHIMNEY (square inches)							
	12	19	28	38	50	63	78	113
6	25	46	71	103	143	188	246	NA
8	28	53	82	119	163	218	278	408
10	31	56	90	131	177	236	302	454
15	NA	67	106	152	212	283	365	546
20	NA	NA	NA	NA	NA	325	419	648
30	NA	NA	NA	NA	NA	NA	496	749
50	NA	NA	NA	NA	NA	NA	NA	922
100	NA	NA	NA	NA	NA	NA	NA	NA

TABLE 504.3(6b)
EXTERIOR MASONRY CHIMNEY

Number of Appliances	Two or more
Appliance Type	NAT + NAT
Appliance Vent Connection	Type B double-wall connector

**Minimum Allowable Input Rating of
Space-heating Appliance in Thousands of Btu per Hour**

VENT HEIGHT (feet)	INTERNAL AREA OF CHIMNEY (square inches)							
	12	19	28	38	50	63	78	113

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37°F or Greater Local 99% Winter Design Temperature: 37°F or Greater								
6	0	0	0	0	0	0	0	NA
8	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0
15	NA	0	0	0	0	0	0	0
20	NA	NA	NA	NA	NA	184	0	0
30	NA	NA	NA	NA	NA	393	334	0
50	NA	NA	NA	NA	NA	NA	NA	579
100	NA	NA	NA	NA	NA	NA	NA	NA
27 to 36°F Local 99% Winter Design Temperature: 27 to 36°F								
6	0	0	68	NA	NA	180	212	NA
8	0	0	82	NA	NA	187	214	263
10	0	51	NA	NA	NA	201	225	265
15	NA	NA	NA	NA	NA	253	274	305
20	NA	NA	NA	NA	NA	307	330	362
30	NA	NA	NA	NA	NA	NA	445	485
50	NA	NA	NA	NA	NA	NA	NA	763
100	NA	NA	NA	NA	NA	NA	NA	NA

(continued)

TABLE 504.3(6b)—continued
EXTERIOR MASONRY CHIMNEY
Minimum Allowable Input Rating of
Space-heating Appliance in Thousands of Btu per Hour

VENT HEIGHT (feet)	INTERNAL AREA OF CHIMNEY (square inches)							
	12	19	28	38	50	63	78	113
17 to 26°F Local 99% Winter Design Temperature: 17 to 26°F								
6	NA	NA	NA	NA	NA	NA	NA	NA
8	NA	NA	NA	NA	NA	NA	264	352
10	NA	NA	NA	NA	NA	NA	278	358
15	NA	NA	NA	NA	NA	NA	331	398
20	NA	NA	NA	NA	NA	NA	387	457
30	NA	NA	NA	NA	NA	NA	NA	581
50	NA	NA	NA	NA	NA	NA	NA	862
100	NA	NA	NA	NA	NA	NA	NA	NA
5°F to 16°F Local 99% Winter Design Temperature: 5°F to 16°F								
6	NA	NA	NA	NA	NA	NA	NA	NA
8	NA	NA	NA	NA	NA	NA	NA	NA

10	NA	NA	NA	NA	NA	NA	NA	430
15	NA	NA	NA	NA	NA	NA	NA	485
20	NA	NA	NA	NA	NA	NA	NA	547
30	NA	NA	NA	NA	NA	NA	NA	682
50	NA	NA	NA	NA	NA	NA	NA	NA
100	NA	NA	NA	NA	NA	NA	NA	NA
4°F or Lower Local 99% Winter Design Temperature: 4°F or Lower								
Not recommended for any vent configurations								

For SI: $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$, 1 inch = 25.4 mm, 1 square inch = 615.16 mm²,
1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

TABLE 504.3(7a)
EXTERIOR MASONRY CHIMNEY

Number of Appliances	Two or more
Appliance Type	FAN + NAT
Appliance Vent Connection	Type B double-wall connector

**Combined Appliance Maximum
Input Rating in Thousands of Btu per Hour**

[illegible]

TABLE 504.3(7b)
EXTERIOR MASONRY CHIMNEY

Number of Appliances	Two or more
Appliance Type	FAN + NAT
Appliance Vent Connection	Type B double-wall connector

**Minimum Allowable Input Rating of
Space-heating Appliance in Thousands of Btu per Hour**

Space-Heating Appliance in Thousands of Btu per Hour								
VENT HEIGHT (feet)	INTERNAL AREA OF CHIMNEY (square inches)							
	12	19	28	38	50	63	78	113
37°F or Greater	Local 99% Winter Design Temperature: 37°F or Greater							

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6	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0
15	NA	0	0	0	0	0	0	0
20	NA	NA	123	190	249	184	0	0
30	NA	NA	NA	334	398	393	334	0
50	NA	NA	NA	NA	NA	714	707	579
100	NA	NA	NA	NA	NA	NA	NA	1,600
27°F to 36°F Local 99% Winter Design Temperature: 27°F to 36°F								
6	0	0	68	116	156	180	212	266
8	0	0	82	127	167	187	214	263
10	0	51	97	141	183	201	225	265
15	NA	111	142	183	233	253	274	305
20	NA	NA	187	230	284	307	330	362
30	NA	NA	NA	330	319	419	445	485
50	NA	NA	NA	NA	NA	672	705	763
100	NA	NA	NA	NA	NA	NA	NA	1,554

(continued)

TABLE 504.3(7b)—continued
EXTERIOR MASONRY CHIMNEY
Minimum Allowable Input Rating of
Space-heating Appliance in Thousands of Btu per Hour

VENT HEIGHT (feet)	INTERNAL AREA OF CHIMNEY (square inches)							
	12	19	28	38	50	63	78	113
17°F to 26°F Local 99% Winter Design Temperature: 17°F to 26°F								
6	0	55	99	141	182	215	259	349
8	52	74	111	154	197	226	264	352
10	NA	90	125	169	214	245	278	358
15	NA	NA	167	212	263	296	331	398
20	NA	NA	212	258	316	352	387	457
30	NA	NA	NA	362	429	470	507	581
50	NA	NA	NA	NA	NA	723	766	862
100	NA	NA	NA	NA	NA	NA	NA	1,669
5°F to 16°F Local 99% Winter Design Temperature: 5°F to 16°F								
6	NA	78	121	166	214	252	301	416
8	NA	94	135	182	230	269	312	423
10	NA	111	149	198	250	289	331	430

15	NA	NA	193	247	305	346	393	485
20	NA	NA	NA	293	360	408	450	547
30	NA	NA	NA	377	450	531	580	682
50	NA	NA	NA	NA	NA	797	853	972
100	NA	NA	NA	NA	NA	NA	NA	1,833
-10°F to 4°F		Local 99% Winter Design Temperature: -10°F to 4°F						
6	NA	NA	145	196	249	296	349	484
8	NA	NA	159	213	269	320	371	494
10	NA	NA	175	231	292	339	397	513
15	NA	NA	NA	283	351	404	457	586
20	NA	NA	NA	333	408	468	528	650
30	NA	NA	NA	NA	NA	603	667	805
50	NA	NA	NA	NA	NA	NA	955	1,003
100	NA	NA	NA	NA	NA	NA	NA	NA
-11°F or Lower		Local 99% Winter Design Temperature: -11°F or Lower						
Not recommended for any vent configurations								

For SI: °C = (°F - 32)/1.8, 1 inch = 25.4 mm, 1 square inch = 615.16 mm²,
1 foot = 304.8 mm, 1 British thermal unit per hour = 0.2931 W.

504.3.1 Vent obstructions. These venting tables shall not be used where obstructions, as described in Section 503.15, are installed in the venting system. The installation of vents serving listed appliances with vent dampers shall be in accordance with the appliance manufacturer's instructions or in accordance with the following:

1. The maximum capacity of the vent connector shall be determined using the NAT Max column.
2. The maximum capacity of the vertical vent or chimney shall be determined using the FAN+NAT column when the second appliance is a fan-assisted appliance, or the NAT+NAT column when the second appliance is equipped with a draft hood.
3. The minimum capacity shall be determined as if the appliance were a fan-assisted appliance.
 - 3.1. The minimum capacity of the vent connector shall be determined using the FAN Min column.
 - 3.2. The FAN+FAN column shall be used where the second appliance is a fan-assisted appliance, and the FAN+NAT column shall be used where the second appliance is equipped with a draft hood, to determine whether the vertical vent or chimney configuration is not permitted (NA). Where the vent configuration is NA, the vent configuration shall not be permitted and an alternative venting configuration shall be utilized.

504.3.2 Connector length limit. The vent connector shall be routed to the vent utilizing the shortest possible route. Except as provided in Section 504.3.3, the maximum vent connector horizontal length shall be 1½ feet for each inch (18 mm for each mm) of connector diameter as shown in Table 504.3.2.

**TABLE 504.3.2
MAXIMUM VENT CONNECTOR LENGTH**

CONNECTOR DIAMETER (inches)	CONNECTOR MAXIMUM HORIZONTAL LENGTH (feet)
3	4½
4	6

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5	7½
6	9
7	10½
8	12
9	13½
10	15
12	18
14	21
16	24
18	27
20	30
22	33
24	36

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

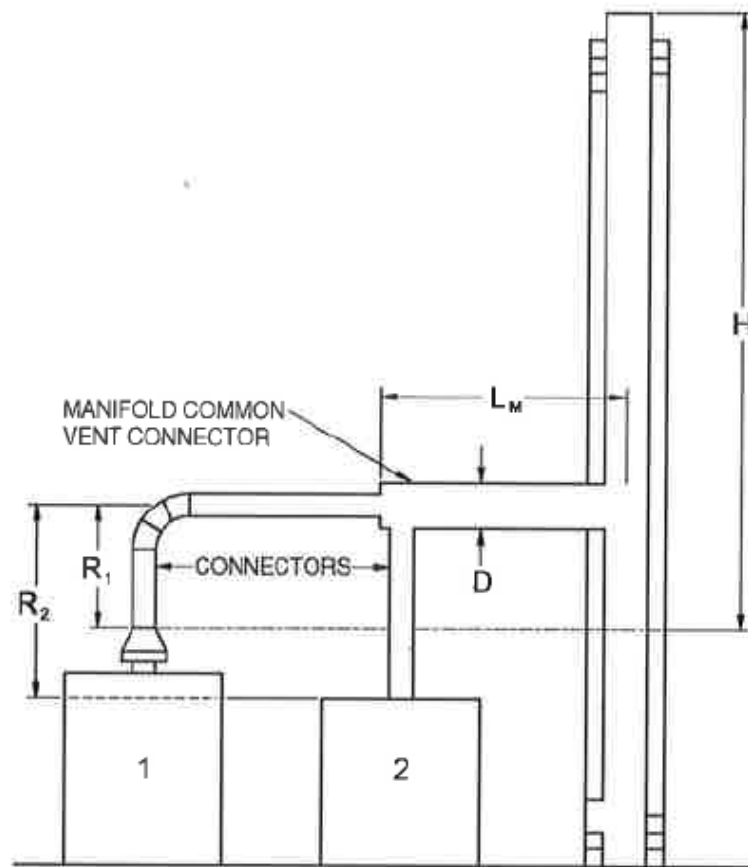
504.3.3 Connectors with longer lengths. Connectors with longer horizontal lengths than those listed in Section 504.3.2 are permitted under the following conditions:

1. The maximum capacity (FAN Max or NAT Max) of the vent connector shall be reduced 10 percent for each additional multiple of the length allowed by Section 504.3.2. For example, the maximum length listed in Table 504.3.2 for a 4-inch (101.6 mm) connector is 6 feet (1829 mm). With a connector length greater than 6 feet (1828.8 mm) but not exceeding 12 feet (3657.6 mm), the maximum capacity must be reduced by 10 percent ($0.90 \times$ maximum vent connector capacity). With a connector length greater than 12 feet (3657.6 mm) but not exceeding 18 feet (5486 mm), the maximum capacity must be reduced by 20 percent ($0.80 \times$ maximum vent capacity).
2. For a connector serving a fan-assisted appliance, the minimum capacity (FAN Min) of the connector shall be determined by referring to the corresponding single appliance table. For Type B double-wall connectors, Table 504.2(1) shall be used. For single-wall connectors, Table 504.2(2) shall be used. The height (H) and lateral (L) shall be measured according to the procedures for a single-appliance vent, as if the other appliances were not present.

504.3.4 Vent connector manifold. Where the vent connectors are combined prior to entering the vertical portion of the common vent to form a common vent manifold, the size of the common vent manifold and the common vent shall be determined by applying a 10 percent reduction ($0.90 \times$ maximum common vent capacity) to the common vent capacity part of the common vent tables. The length of the common vent connector manifold (Lm) shall not exceed 1½ feet for each inch (18 mm for each mm) of common vent connector manifold diameter (D) (see Figure 504.3.4).

504.3.5 Common vertical vent offset. Where the common vertical vent is offset, the maximum capacity of the common vent shall be reduced in accordance with Section 504.3.6. The horizontal length of the common vent offset (Lo) shall not exceed 1½ feet for each inch (18 mm for each mm) of common vent diameter (D). Where multiple offsets occur in a common vent, the total horizontal length of all offsets combined shall not exceed 1½ feet for each inch (18 mm for each mm) of common vent diameter (D).

504.3.6 Elbows in vents. For each elbow up to and including 45 degrees (0.79 rad) in the common vent, the maximum common vent capacity listed in the venting tables shall be reduced by 5 percent. For each elbow greater than 45 degrees (0.79 rad) up to and including 90 degrees (1.57 rad), the maximum common vent capacity listed in the venting tables shall be reduced by 10 percent.



Example: Manifolded Common Vent Connector L_M shall be no greater than 18 times the common vent connector inside diameter; i.e., a 4-inch (102 mm) inside diameter common vent connector manifold shall not exceed 72 inches (1829 mm) in length (see Section 504.3.4).

Note: This is an illustration of a typical manifolded vent connector. Different appliance, vent connector, or common vent types are possible. Consult Section 502.3.

FIGURE 504.3.4
USE OF A MANIFOLD COMMON VENT CONNECTOR

504.3.7 Elbows in connectors. The vent connector capacities listed in the common vent sizing tables include allowance for two 90-degree (1.57 rad) elbows. For each additional elbow up to and including 45 degrees (0.79 rad), the maximum vent connector capacity listed in the venting tables shall be reduced by 5 percent. For each elbow greater than 45 degrees (0.79 rad) up to and including 90 degrees (1.57 rad), the maximum vent connector capacity listed in the venting tables shall be reduced by 10 percent.

504.3.8 Common vent minimum size. The cross-sectional area of the common vent shall be equal to or greater than the cross-sectional area of the largest connector.

504.3.9 Common vent fittings. At the point where tee or wye fittings connect to a common vent, the opening size of the fitting shall be equal to the size of the common vent. Such fittings shall not be prohibited from having reduced size openings at the point of connection of appliance vent connectors.

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504.3.9.1 Tee and wye fittings. Tee and wye fittings connected to a common gas vent shall be considered as part of the common gas vent and shall be constructed of materials consistent with that of the common gas vent.

504.3.10 Reserved.

504.3.11 Connector rise measurement. Connector rise (R) for each appliance connector shall be measured from the draft hood outlet or flue collar to the centerline where the vent gas streams come together.

504.3.12 Vent height measurement. For multiple units of appliances all located on one floor, available total height (H) shall be measured from the highest draft hood outlet or flue collar up to the level of the outlet of the common vent.

504.3.13 Reserved.

504.3.14 Reserved.

504.3.15 Reserved.

504.3.16 Reserved.

504.3.17 Vertical vent maximum size. Where two or more appliances are connected to a vertical vent or chimney, the flow area of the largest section of vertical vent or chimney shall not exceed seven times the smallest listed appliance categorized vent areas, flue collar area or draft hood outlet area unless designed in accordance with approved engineering methods.

504.3.18 Multiple input rate appliances. For appliances with more than one input rate, the minimum vent connector capacity (FAN Min) determined from the tables shall be less than the lowest appliance input rating, and the maximum vent connector capacity (FAN Max or NAT Max) determined from the tables shall be greater than the highest appliance input rating.

504.3.19 Liner system sizing and connections. Listed, corrugated metallic chimney liner systems in masonry chimneys shall be sized by using Table 504.3(1) or 504.3(2) for Type B vents, with the maximum capacity reduced by 20 percent ($0.80 \times$ maximum capacity) and the minimum capacity as shown in Table 504.3(1) or 504.3(2). Corrugated metallic liner systems installed with bends or offsets shall have their maximum capacity further reduced in accordance with Sections 504.3.5 and 504.3.6. The 20-percent reduction for corrugated metallic chimney liner systems includes an allowance for one long-radius 90-degree (1.57 rad) turn at the bottom of the liner. Where double-wall connectors are required, tee and wye fittings used to connect to the common vent chimney liner shall be listed double-wall fittings. Connections between chimney liners and listed double-wall fittings shall be made with listed adapter fittings designed for such purpose.

504.3.20 Chimney and vent location. Tables 504.3(1), 504.3(2), 504.3(3), 504.3(4), and 504.3(5) shall be used only for chimneys and vents not exposed to the outdoors below the roof line. Tables 504.3(6a), 504.3(6b), 504.3(7a) and 504.3(7b) shall be used for clay-tile-lined exterior masonry chimneys, provided that all of the following conditions are met:

1. Vent connectors are Type B double wall.
2. Not less than one appliance is draft hood equipped.
3. The combined appliance input rating is less than the maximum capacity given by Table 504.3(6a) for NAT+NAT or Table 504.3(7a) for FAN+NAT.
4. The input rating of each space-heating appliance is greater than the minimum input rating given by Table 504.3(6b) for NAT+NAT or Table 504.3(7b) for FAN+NAT.
5. The vent connector sizing is in accordance with Table 504.3(3).

504.3.21 Connector maximum and minimum size. Vent connectors shall not be increased in size more than two sizes greater than the listed appliance categorized vent diameter, flue collar diameter or draft hood outlet diameter. Vent connectors for draft hood-equipped appliances shall not be smaller than the draft hood outlet diameter. Where a vent connector size(s) determined from the tables for a fan-assisted appliance(s) is smaller than the flue collar diameter, the use of the smaller size(s) shall be permitted provided that the installation complies with all of the following conditions:

1. Vent connectors for fan-assisted appliance flue collars 12 inches (304.8 mm) in diameter or smaller are not reduced by more than one table size (e.g., 12 inches to 10 inches (304.8 mm to 254 mm) is a one-size reduction) and those larger than 12 inches (304.8 mm) in diameter are not reduced more than two table sizes (e.g., 24 inches to 20 inches (609.6 mm to 508 mm) is a two-size reduction).
2. The fan-assisted appliance(s) is common vented with a draft-hood-equipped appliance(s).
3. The vent connector has a smooth interior wall.

504.3.22 Component commingling. All combinations of pipe sizes, single-wall and double-wall metal pipe shall be allowed within any connector run(s) or within the common vent, provided that all of the appropriate tables permit all of the desired sizes and types of pipe, as if they were used for the entire length of the subject connector or vent. Where single-wall and Type B double-wall metal pipes are used for vent connectors within the same venting system, the common vent must be sized using Table 504.3(2) or 504.3(4), as appropriate.

504.3.23 Draft hood conversion accessories. Draft hood conversion accessories for use with masonry chimneys venting listed Category I fan-assisted appliances shall be listed and installed in accordance with the manufacturer's instructions for such listed accessories.

504.3.24 Multiple sizes permitted. Where a table permits more than one diameter of pipe to be used for a connector or vent, all the permitted sizes shall be permitted to be used.

504.3.25 Table interpolation. Interpolation shall be permitted in calculating capacities for vent dimensions that fall between table entries.

504.3.26 Extrapolation prohibited. Extrapolation beyond the table entries shall not be permitted.

504.3.27 Engineering calculations. For vent heights less than 6 feet (1828.8 mm) and greater than shown in the tables, engineering methods shall be used to calculate vent capacities.

504.3.28 Height entries. Where the actual height of a vent falls between entries in the height column of the applicable table in Tables 504.3(1) through 504.3(7b), either interpolation shall be used or the lower appliance input rating shown in the table shall be used for FAN MAX and NAT MAX column values and the higher appliance input rating shall be used for the FAN MIN column values.

SECTION FGC 505

DIRECT-VENT, INTEGRAL VENT, MECHANICAL VENT AND VENTILATION/EXHAUST HOOD VENTING

505.1 General. The installation of direct-vent and integral vent appliances shall be in accordance with Section 503. Mechanical venting systems and exhaust hood venting systems shall be designed and installed in accordance with Section 503.

505.1.1 Commercial cooking appliances vented by exhaust hoods. Refer to Chapter 5 of the *New York City Mechanical Code*.

505.1.2 Interlock requirements. Where commercial cooking appliances are vented by means of the Type I or Type II kitchen exhaust hood system that serves such appliances, the exhaust system shall be fan powered and the appliances shall be interlocked with the exhaust hood system to prevent appliance operation when the exhaust hood system is not operating. The method of interlock between the exhaust hood system and the appliances equipped with standing pilot burner ignition systems shall not cause such pilots to be extinguished. Where a solenoid valve is installed in the gas piping as part of an interlock system, gas piping shall not be installed to bypass such valve. Dampers shall not be installed in the exhaust system.

Exception: An interlock between the cooking appliance(s) and the exhaust hood system shall not be required where heat sensors or other approved methods automatically activate the exhaust hood system when cooking operations occur.

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SECTION FGC 506 FACTORY-BUILT CHIMNEYS

506.1 Building heating appliances. Factory-built chimneys for building heating appliances producing flue gases having a temperature not greater than 1,000°F (537.8°C), measured at the entrance to the chimney, shall be listed and labeled in accordance with UL 103 and shall be installed and terminated in accordance with the manufacturer's instructions and this code.

506.2 Support. Where factory-built chimneys are supported by structural members, such as joists and rafters, such members shall be designed to support the additional load.

506.3 Medium-heat appliances. Factory-built chimneys for medium-heat appliances producing flue gases having a temperature above 1,000°F (537.8°C), measured at the entrance to the chimney, shall be listed and labeled in accordance with UL 959 and shall be installed and terminated in accordance with the manufacturer's instructions and this code.

SECTION FGC 507 CHANGES IN APPLIANCE FUELS

507.1 Changes in appliance fuels. Conversion of appliances from solid or liquid fuel to natural gas or the addition of natural gas, shall be made in accordance with this code. Conversion from natural gas to, or the addition of, #2 fuel oil for a heating appliance shall be made only if:

1. The chimney design meets the requirements of this chapter for the conversion fuel and the chimney is test run and smoke tested in accordance with Sections 503.5.6.4 and 503.5.6.5.
2. The chimney is sized to provide adequate draft and to vent the combustion products for the new fuel.
3. The chimney is thoroughly cleaned prior to the conversion to remove collected flue deposits.

SECTION FGC 508 EXHAUST GASES FROM INTERNAL COMBUSTION ENGINES AND TURBINES

508.1 Exhaust system construction. The exhaust system from internal combustion engines and turbines shall be constructed in accordance with Section 811 of the *New York City Mechanical Code*.